

Spectral Energy is Not a Reliable Collapse Diagnostic for Low-Resource Quantized LoRA: Evidence from Turkic Languages

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Abstract

Spectral energy—the concentration of an adapter’s singular-value spectrum—has been proposed as a weight-based diagnostic of low-rank adaptation (LoRA) fine-tuning failure, validated on English at scale. We test whether it remains reliable where cheap diagnostics are needed most: low-resource, morphologically complex languages, for which full evaluation suites rarely exist. Fine-tuning Gemma-2-9B with QLoRA on Kyrgyz, Kazakh, and Uzbek, with a Llama-3-8B replication, we report four findings. **(1)** The spectral-energy threshold silently misses learning-rate-induced collapse: runs with near-identical, sub-threshold spectra span the full functional range from catastrophic collapse (zero NER F1, near-random QA) to healthy behaviour. A controlled decomposition traces the collapse to learning rate, shows that the apparent benefit of higher rank is an artefact of the common $\alpha=2r$ convention silently scaling the per-step update, and rules out 4-bit quantization with a full-precision control. **(2)** Frobenius norm growth separates regimes within a language category but fails across categories: growth that leaves English unharmed accompanies severe forgetting for Kazakh. **(3)** The final adapter is structurally underdetermined: across random seeds, direct LoRA lands in near-orthogonal directions, and the collapse-vs-healthy directional gap lies inside this seed-noise band—within the regime we measure, no Lipschitz function of the final adapter alone can separate the two. Detection must come from functional or trajectory probes. **(4)** Related-language warm-start (Kazakh→Kyrgyz) robustly halves cross-lingual forgetting on both architectures, while a cross-family (English) warm-start does not. Directionally, transfer pins the adapter to its source initialization, but transfers from independent sources are again near-orthogonal—extending the underdetermination result to transfer. We release all training, evaluation, and SVD-monitoring code, the SVD logs, and the per-run evaluation reports

for every configuration.¹

1 Introduction

Low-Rank Adaptation (LoRA; Hu et al., 2022) fine-tunes a large language model by learning an additive update $\Delta W = BA$ to each frozen weight matrix, where A and B are trainable factors of rank $r \ll \min(d, k)$ and a scaling hyperparameter α sets the update’s magnitude; it is the standard tool for compute-constrained fine-tuning. Biderman et al. (2024) proposed *spectral collapse*—the concentration of the update’s singular values σ_i , measured by spectral energy $SE = \sigma_1 / \sum_i \sigma_i$ exceeding ~ 0.7 —as the diagnostic of LoRA degradation, established on English at large scale.

Whether such weight-based diagnostics transfer to low-resource settings is where the answer matters most. Practitioners adapting LLMs to underrepresented languages rarely have the evaluation suites needed to functionally test every checkpoint; a health signal computed directly from the adapter weights would be the natural substitute. Yet this is also the hardest case: adapters must make larger perturbations to accommodate poorly covered, morphologically complex languages, and whether thresholds tuned on English survive there has not been systematically tested. Our research question: *can any function of the final adapter’s weights be trusted to detect fine-tuning failure in this regime?*

We test three hypotheses on Gemma-2-9B and replicate the central claims on Llama-3-8B, using Kyrgyz, Kazakh, and Uzbek as a stress test ($3.2\times$ subword fragmentation under standard tokenizers, severe pretraining underrepresentation). **H1** (SE threshold): $SE > 0.7$ predicts functional degradation. **H2** (transfer): related-language warm-start preserves source-language knowledge at minimal

¹<https://anonymous.4open.science/r/Spectral-Collapse-Turkic-Conference-6DE4>

Frobenius perturbation. **H3** (directional discriminator): some scalar geometric function of the final adapter separates collapse from healthy training at the same rank.

Contributions. (C1) A controlled four-way decomposition (E5/E5b/E5c plus a new $\alpha=32$ control C5) **falsifies the SE-threshold hypothesis** for low-resource quantized LoRA: four $r=64$ runs at sub-threshold SE span the full functional range from catastrophic collapse to behaviour matching the $r=16$ baseline; SE cannot distinguish them. C3 (BF16) rules out quantization; C5 shows the rank-only effect of E5c is an artefact of the $\alpha=2r$ convention amplifying absolute α at higher rank, not of rank itself (Sections 4.1 and 4.3). (C2) Frobenius growth is a useful but language-dependent diagnostic; it fails across language categories (EN $33\times$ no forgetting vs. KZ $38\times$ severe forgetting; Section 4.4). (C3) We **falsify H3 and derive a structural underdetermination argument** (Section 5.2): a 22-pair pairwise-cosine sweep shows direct LoRA from base is directionally non-canonical (cross-seed cosine 0.01–0.04, 0/294 modules above 0.5); the collapse-vs-healthy gap sits inside this seed-noise band, so within the measured regime no Lipschitz scalar function of the final adapter alone can separate the two. (C4) Related-language warm-start **preserves source-language knowledge**: KZ→KY transfer halves cross-lingual KZ PPL across two seeds (23.5 ± 0.1 vs. 50.0 ± 2.4 , +13% Frobenius growth) on Gemma, replicates on Llama-3 ($30.4 \rightarrow 19.0$), and survives architecturally-fair independent-init reseeding ($50.0 \rightarrow 26.79$, –47% retention). The cross-family control (EN→KY) does not reproduce retention. Transfer pins the adapter to its source initialization ($\cos \approx 0.80$ on both architectures); two transfers from *independent* sources, however, land in orthogonal directions ($\cos$ 0.023 Gemma, 0.060 Llama-3) – the underdetermination argument extends to transfer once the source itself is reseeded. (C5) A Llama-3-8B replication across five configurations (E3, E5c, E5 seed-replicated; plus C5 absolute- α control and a two-seed independent E6 KZ→KY transfer) reproduces all four findings, including the refined source-pinning analysis of (C4) (Section T). (C6) We release a pairwise-cosine + SVD-monitoring toolkit and all training/eval code.

2 Related Work

Weight-based health diagnostics. A fine-tuning run can fail in ways that only a full evaluation suite reveals—but running one on every checkpoint is expensive, and for many languages the suites do not exist. This motivated a line of work reading training health directly off the weights, which is cheap and needs no task data. Martin and Mahoney (2021) showed that the singular-value spectra of well-trained matrices carry a signature of implicit self-regularisation, establishing spectra as a proxy for training quality. In the LoRA setting, Biderman et al. (2024) turned this into an operational rule: as an adapter degrades, its update concentrates into a single direction, so *spectral energy* $SE = \sigma_1 / \sum_i \sigma_i$ rises, and $SE > 0.7$ flags collapse. Jang et al. (2024) analysed LoRA’s optimisation trajectory in the NTK regime. Both establish their results on English at large scale; whether the threshold survives elsewhere is what we test.

LoRA and its variants. LoRA (Hu et al., 2022) learns $\Delta W = BA$ at rank r ; QLoRA (Dettmers et al., 2023) adds a 4-bit quantised base so 9B-parameter models fit on consumer GPUs. Several variants change *where* the adapter starts or *how* it moves: rank allocation (Zhang et al., 2023), weight-decomposed updates (DoRA, Liu et al., 2024), singular-value-aligned initialisation (PiSSA, Meng et al., 2024), and per-factor LR rescaling (LoRA+, Hayou et al., 2024). That fine-tuning succeeds at low rank at all is consistent with its low intrinsic dimension (Aghajanyan et al., 2021)—a fact we return to in Section 5.2, since a low-dimensional solution space says nothing about *which* low-dimensional subspace a given run selects.

Forgetting under adaptation. Adapting a model to a new language degrades the old ones (McCloskey and Cohen, 1989; Kirkpatrick et al., 2017), and low-rank constraints do not prevent it (Luo et al., 2023; Kotha et al., 2024; Kumar et al., 2022); severity grows with distance from the pre-training distribution (Csaki et al., 2024). Our sequential transfer construction is closest in spirit to MAD-X (Pfeiffer et al., 2020), but warm-starts a single adapter rather than composing parallel language adapters.

Turkic NLP. Standardised evaluation is recent: TUMLU (Isbarov et al., 2025) covers multi-task understanding across Turkic languages, and tokenizer mismatch for agglutinative morphology is

quantified across model families (Rust et al., 2021; Ahia et al., 2023)—the fragmentation that makes these languages a stress test for adaptation.

This work. Weight-based diagnostics are validated where evaluation is cheap and applied where it is not. We test whether that transfer is warranted, and find that it is not: the relationship between final-adapter geometry and functional failure in multilingual LoRA has not, to our knowledge, been systematically tested, and turns out to be weaker than the diagnostics assume.

3 Experimental Setup

3.1 Base Model

We use Gemma-2-9B (Gemma Team et al., 2024) as our base model, a 9.24 billion parameter decoder-only transformer with 42 layers (per the official model configuration).² The model is quantized to 4-bit precision using the NF4 data type with double quantization (Dettmers et al., 2023), reducing memory requirements to approximately 5.1 billion effective parameters while preserving model quality. All experiments use bfloat16 compute precision. We chose Gemma-2-9B because it possesses baseline linguistic knowledge of Turkic languages from pretraining, allowing us to isolate the spectral dynamics of LoRA *adaptation* from the process of primary language acquisition. The functional collapse observed at $r=64$ therefore reflects a pathology of the adaptation process, not an inability of the model to represent Turkic tokens.

3.2 Training Data

We compile monolingual corpora of approximately 150 MB for each of the three target languages:

- **Kazakh (KZ):** 61,879 records sourced from Kazakh Wikipedia and the sozkz-corpus, totaling 14.1M tokens (151.5 MB).
- **Kyrgyz (KY):** 17,360 records from a curated local corpus comprising literature, history, and encyclopedic texts, totaling 4.4M tokens (151.4 MB).³
- **Uzbek (UZ):** 21,242 records from uz-books, Uzbek Wikipedia, and FineWeb-2 with Cyrillic script filtering, totaling 5.4M tokens

²<https://huggingface.co/google/gemma-2-9b/blob/main/config.json> reports `num_hidden_layers=42`.

³Due to licensing restrictions on the source materials, the Kyrgyz corpus cannot be publicly redistributed. Corpus statistics and preprocessing details are provided in Section E.

(152.6 MB). Note that our Uzbek corpus uses Cyrillic script, which reflects a subset of Uzbek digital content; results may not generalize to Latin-script Uzbek, which dominates contemporary formal writing.

Additionally, we compile an **English (EN)** control corpus of 52,268 records from English Wikipedia (155 MB, 12.7M tokens), matched in size to the Turkic corpora.

All texts are chunked into segments of 200–5,000 characters. A 10% held-out split is reserved for validation. All corpora are matched by raw byte size (~ 150 MB), but under comparable subword fragmentation ($\sim 3.2 \times$ tokens per word across all three languages; Section 3.6) token counts still differ substantially (KZ: 14.1M, UZ: 5.4M, KY: 4.4M). This reflects a *source-text difference*—the Kazakh corpus is dominated by dense encyclopedic prose with shorter word forms, while the Kyrgyz corpus contains a higher proportion of literary material with longer word forms. Because tokenizer fragmentation is nearly identical across the three languages, this imbalance is a property of the corpora, not of the tokenizer. We flag it explicitly as a confound for any direct E1 vs. E3 comparison (see also Limitation L5): Kazakh’s superior target PPL could partially reflect $3 \times$ more training tokens rather than better tokenizer coverage for Kazakh per se. We control for this with E1b, a reduced-Kazakh subset of ~ 1.5 M training tokens—roughly $3 \times$ *smaller* than the KY baseline (4.4M tokens). The original goal was a token-matched control, but the corpus subsetting (sized via $\text{word} \times \text{fragmentation-ratio}$ estimation) under-shot once actual chunked tokens were counted; the comparison still serves its purpose, as discussed in Sections 4.2 and 6.

3.3 LoRA Configuration

We apply LoRA to all attention and MLP projection matrices (q, k, v, o , gate, up, down — 7 modules per layer, 294 total across 42 Gemma-2 layers; 224 across 32 Llama-3 layers). $\alpha = 2r$ throughout; we use paged AdamW with cosine LR schedule and 5% warmup; effective batch 16; max seq length 256. The complete configuration grid (12 Gemma-2 runs, 5 Llama-3 runs, all language/rank/LR/epochs/init combinations) is in Table 6 (Section E). Ten of twelve Gemma-2 configurations are seed-replicated at $n=2$ (E1, E1b, E2, E3, E5, E5b, E5c, E6, E6b, E8; see Section J); on

Llama-3, E3, E5c, and E5 are replicated at $n=2$.

3.4 SVD Monitoring Protocol

We SVD each `lora_B` matrix every 100 training steps and log five spectral metrics: spectral energy $SE = \sigma_1 / \sum_i \sigma_i$ (Biderman et al., 2024), effective rank $\exp(-\sum_i p_i \log p_i)$ with $p_i = \sigma_i / \sum_j \sigma_j$ (Roy and Vetterli, 2007), stable rank $\|W\|_F^2 / \sigma_1^2$, SVD entropy, and Frobenius norm. We monitor A and B separately. The SE-collapse threshold of 0.7 from Biderman et al. (2024) is the principal diagnostic we test. The pairwise-cosine analysis (Section 4.6) is computed on the full $\Delta W = BA$ matrix.

3.5 Evaluation Protocol

We evaluate each adapter on three tasks:

Perplexity (PPL): Computed on held-out validation splits (500 samples, 113–128K tokens per language). We construct a *cross-lingual perplexity matrix* by evaluating each adapter on all three languages, providing a direct measure of catastrophic forgetting.

Named Entity Recognition (NER): Using the WikiANN dataset (Pan et al., 2017) with 3-shot prompting on 100 test samples per language. We report two complementary metrics: (i) generation-based F1 micro-averaged across PER/ORG/LOC types (subject to parse-failure noise and ± 0.085 Wilson CI at $n=100$); and (ii) log-likelihood span typing accuracy on gold-span boundaries (Section G)—a parse-failure-immune probe that scores entity-type knowledge directly. The log-likelihood metric is our primary NER number in Table 2; F1 is reported alongside for comparability with prior work. **Caveat for Uzbek:** our UZ corpus uses Cyrillic script while WikiANN-uz is primarily Latin (overlap 2%, see L9 and Section H); cross-configuration UZ NER comparisons should therefore be read as cross-script transfer rather than in-distribution NER.

Question Answering (QA): Using TUMLU (Isbarov et al., 2025), a multiple-choice benchmark for Turkic language understanding, with 5-shot log-likelihood evaluation on 695–794 questions per language.

3.6 Subword Tokenization Analysis

We measure subword fragmentation as the average number of tokens per whitespace-delimited word for each language. As shown in Table 1, all three Turkic languages exhibit severe fragmentation at approximately $3.2\times$ the rate of English, reflecting

the fundamental mismatch between the Gemma tokenizer (optimized for English and high-resource languages) and agglutinative Turkic morphology.

Language	Tokens/Word	Ratio to EN
English	1.15	1.00×
Uzbek	3.68	3.20×
Kyrgyz	3.69	3.21×
Kazakh	3.74	3.25×

Table 1: Subword fragmentation ratios for the Gemma-2-9B tokenizer. Turkic languages require approximately $3.2\times$ more tokens per word than English, reflecting severe tokenizer mismatch for agglutinative morphology.

4 Results

4.1 Baseline Performance

Spectral collapse versus functional collapse. A central conceptual distinction in our work is between two notions of "collapse" that the LoRA literature has tended to conflate. *Spectral collapse* refers to a geometric property of the adapter weights: the singular value distribution concentrating into a single dominant direction (high SE). *Functional collapse* refers to a behavioural property of the resulting model: loss of target-task capability such as instruction-following or coherent generation (NER F1=0, TUMLU near random, degenerate-loop output; concrete examples in Section N). The SE-threshold diagnostic (Biderman et al., 2024) implicitly assumes the two co-occur. Our results show they can dissociate: E5 exhibits clear functional collapse with sub-threshold SE, and E5c shows healthy spectral and functional trajectories at higher rank. The motivation for tracking Frobenius growth + cross-lingual PPL probes (Section 4.4) is precisely to capture functional collapse signals that pure spectral shape misses.

Table 2 presents the primary evaluation metrics across all experiments. Several patterns emerge from the baseline comparisons.

Target-language PPL tracks training-corpus token count and pretraining coverage (E1 KZ 2.73, E2 UZ 4.03, E3 KY 4.78). NER F1 (gen) is noisy at $n=100$ (Wilson ± 0.085); we use log-likelihood TypeAcc on gold spans (Section G) as the primary entity-knowledge metric. The $r=64$ high-LR configuration (E5) is the central empirical surprise: target PPL only 5.90 (+23% over baseline), but NER F1=0 everywhere (100% parse failures), TUMLU collapses to 23.2% (near 25% random). Under log-likelihood TypeAcc, however, E5 retains the

ID	Experiment	Target Lang	PPL (target)	NER TypeAcc (KY)	NER F1 (gen) (KY, ± 0.09 CI)	TUMLU (target)
E1	KZ Baseline (14.1M tok)	KZ	2.73	57.7%	0.219	38.4%
E1b	KZ small-corpus (1.5M tok)	KZ	4.17	46.0%	0.150	35.6%
E2	UZ Baseline	UZ	4.03	54.9%	0.138	35.4%
E3	KY Baseline	KY	4.78	50.4%	0.157	34.7%
E4	KY Overfit (10ep)	KY	4.18	56.8%	0.173	33.9%
E5	KY $r=64$ high-LR	KY	5.90	59.5% [‡]	0.000	23.2%
E5b	KY $r=64$ std-LR (5ep)	KY	4.55	53.1%	0.146	33.2%
E5c	KY $r=64$ epoch-matched (3ep)	KY	3.86	54.9%	0.115	38.5%
E6	KZ→KY	KY	4.73	59.5%	0.253	33.7%
E6b	EN→KY (cross-family)	KY	4.78	54.0%	0.133	36.1%
E8	EN Control	EN	5.54 [*]	56.8%	0.209	37.7% [†]
C3	KY BF16 (no quant)	KY	4.99	58.6%	0.164	36.1%
C5	KY $r=64$, $\alpha=32$ (matched- α ctrl)	KY	4.46	52.3%	0.174	33.2%

Table 2: Main evaluation results across all experiments. **NER TypeAcc (KY)** is our primary NER metric—log-likelihood span typing on gold WikiANN-ky boundaries (Section G); it is parse-failure immune and has lower seed-variance than generation F1. **NER F1 (gen)** is the generation-based score with ± 0.085 Wilson 95% CI at $n=100$, retained for comparability with prior work. We report cross-lingual NER and KZ TUMLU in the appendix (Sections K and L). **E5 vs. E5b/E5c disentangles the rank–LR–epoch confound:** catastrophic NER parse failure at high LR (E5, F1 = 0, TUMLU near random) vanishes at baseline LR (E5b/E5c) and at matched epochs (E5c) outperforms the $r=16$ baseline. **C5 (the $\alpha=32$ control at $r=64$, added to close the absolute- α confound L3) clarifies that E5c’s target-PPL advantage came from the $\alpha=2r$ convention setting $\alpha=128$ at $r=64$ ($4\times$ E3’s $\alpha=32$): at matched absolute α ($\alpha=32$ at both ranks, hence different α/r ratios—2 at $r=16$, 0.5 at $r=64$), the $r=64$ adapter is functionally indistinguishable from the $r=16$ baseline (KY PPL 4.46 vs. 4.78; KZ PPL 49.22 vs. 47.58; cf. E5c 93.5). ^{*}EN PPL on Kazakh; [†]EN TUMLU on Kyrgyz. [‡]E5’s high TypeAcc despite F1 = 0 shows entity knowledge survives the collapse—it is structured-output generation that fails (Section G). **Values shown are for seed 42 (single seed)**; for seed-replicated configurations the corresponding two-seed means with spreads are reported in Table 12 (e.g., E3 KY PPL 4.81 ± 0.02 , E5c 3.89 ± 0.03); small differences between the two views are within seed variance.**

highest KY type accuracy in the sweep (59.5%), so the collapse is loss of structured-output instruction-following, not erasure of entity knowledge.

Rank–LR–epoch– α decomposition (E5 vs. E5b vs. E5c vs. C5). E5 ($r=64$, $\text{lr}=5\times 10^{-4}$, 5 epochs, $\alpha=128$) varies four factors at once. E5b matches rank but uses $\text{lr}=2\times 10^{-4}$ (replicated at seeds 42, 123): collapse vanishes (F1 ≥ 0.12 , TUMLU 33%), so the proximate cause is LR. E5c additionally matches epochs to baseline: at matched rank, LR, epochs, $r=64$ *appears* to outperform the $r=16$ baseline (KY PPL 3.86 vs. 4.78). C5 closes this: holding the absolute α fixed at 32 (same as E3, instead of $\alpha=128$ that the $\alpha=2r$ convention would dictate at $r=64$), $r=64$ is functionally indistinguishable from the $r=16$ baseline—KY PPL 4.46 vs. 4.78, KZ PPL 49.22 vs. 47.58, UZ PPL 57.36 vs. 56.85, F1 0.174 vs. 0.157, TUMLU 33.2% vs. 34.7%—all within seed-spread or Wilson CI. The directional cosine concurs: $\cos(\text{C5}, \text{E3}) = 0.155$ (different rank, same absolute α) is higher than $\cos(\text{C5}, \text{E5c}) = 0.072$ (same rank, different absolute α). Decomposition: catastrophic

collapse = aggressive base LR; cross-lingual forgetting growth = training duration; rank-only effect at matched absolute α = none. The apparent rank advantage of E5c was an artefact of the $\alpha=2r$ convention, which at $r=64$ sets $\alpha=128$ ($4\times$ E3’s $\alpha=32$) and thus amplifies the per-step adapter update; in standard practice, “higher rank” is a proxy for “larger absolute α at fixed optimizer LR” and does not deliver an independent functional benefit at matched α . All four runs satisfy mean SE < 0.1 and final max SE 0.06–0.24, spanning the full functional range; SE never discriminates them.

4.2 Cross-Lingual Degradation

Three patterns drive the rest of the paper (full PPL matrix and heatmap in Section B). **(i)** The $r=64$ high-LR run (E5) shows catastrophic forgetting: KZ PPL 659.25, UZ PPL 742.73 ($\sim 240\times$ over baselines). **(ii)** KZ→KY transfer (E6) halves source-language KZ PPL (23.49 vs. 47.58 for direct KY training) while target KY PPL is unchanged (4.73 vs. 4.78). **(iii)** The EN control (E8) absorbs comparable Frobenius growth ($33\times$ vs. KZ

baseline’s $38\times$) with KZ PPL of only 5.54: same magnitude of perturbation produces qualitatively different cross-lingual outcomes depending on pre-training coverage of the target language. We probe the data-volume vs. pretraining-coverage decomposition (E1 vs. E1b) in Section C: E1b at $\sim 1.5\text{M}$ tokens still attains target KZ PPL 4.17, below the KY baseline’s 4.78 at $\sim 3\times$ more tokens, so coverage is a genuine confound on top of corpus size.

4.3 Spectral Energy Is Not a Reliable Collapse Diagnostic

The SE threshold of 0.7 proposed by [Biderman et al. \(2024\)](#) is never reached in any of our experiments, including those exhibiting severe functional degradation (SE dynamics in Figure 2, layer-wise heatmap in Figure 3, both in Section A). The KY baseline ($r=16$) peaks at max SE ~ 0.44 then decreases to ~ 0.32 . Both $r=64$ configurations—collapsed E5 and healthy E5b—end at final max SE $\sim 0.06\text{--}0.11$, mean SE ~ 0.03 , near-identical spectra but with completely different functional outcomes (F1 0 vs. 0.146, TUMLU 23% vs. 33%). SE is a *shape* metric invariant to rescaling: it cannot see the difference. Effective rank tells the same story: both $r=64$ runs use most of their capacity yet diverge functionally.

Quantization is not the cause (BF16 control, C3). The C3 BF16 replication of E3 (same setup, no quantization) is statistically indistinguishable from 4-bit E3 on the spectral signal we test: final max SE 0.32 vs. 0.33 ± 0.01 , in-training peak 0.47 vs. 0.45 ± 0.02 , Frobenius growth $8.6\times$ vs. $8.4\times$. Target-language KY PPL (4.99 vs. 4.81 ± 0.02), KY NER F1/TypeAcc, and TUMLU sit within or near the two-seed E3 spread. Cross-lingual KZ/UZ PPL under C3 (43.97/59.02) trends slightly outside the E3 spread (50.0 ± 2.4 / 55.5 ± 1.4) but stays in the same regime—an order of magnitude below E5’s catastrophic forgetting and far from the transfer regime—so the spectral falsification carries from 4-bit to full precision regardless. With C3 at $n=1$ a second BF16 seed would tighten the cross-lingual comparison (Section I).

4.4 Frobenius Norm Is Not a Universal Diagnostic

The Frobenius growth ratio $\|B\|_F^{\text{final}}/\|B\|_F^{\text{init}}$ captures scale information SE discards, and separates regimes within an underrepresented-language category ($1.13\times$ transfer, $8.4\times$ direct KY, $15\text{--}38\times$ pathological $r=64$; Table 5, Section D). It fails

across categories: the EN control (E8) has $33\times$ growth—comparable to the KZ baseline’s $38\times$ —but minimal cross-lingual forgetting (KZ PPL 5.52 ± 0.02 across 2 seeds vs. ~ 48 for the KY baseline at $8\times$). E5 and E5b show comparable Frobenius growth ($15.6\times$ vs. $18.2\times$), yet only E5 produces NER parse failure. Norm growth is a reliable diagnostic *within* a language category but not *across*: the same magnitude of perturbation has qualitatively different effects depending on target-language representedness in the base. We take up the directional question in Section 4.6.

4.5 Cross-Lingual Transfer

KZ→KY transfer (E6) initializes LoRA from the final Kazakh adapter (E1) before continuing on Kyrgyz. The robust effect (both seeds 42 and 123) is *source-language retention*: cross-lingual KZ PPL 23.5 ± 0.1 under E6 vs. 50.0 ± 2.4 under direct KY training (E3)—both reported as two-seed mean and half-range; the single-seed-42 numbers in Table 4 (23.49 vs. 47.58) sit inside these ranges—a $2\times$ improvement well outside seed variance, achieved with only $+13\%$ Frobenius growth vs. $8.4\times$ for direct training. Target KY PPL is unchanged (E6 seed-42 4.73 vs. E3 seed-42 4.78; two-seed E6 4.76 ± 0.03 vs. E3 4.81 ± 0.03), and a single-seed KY NER F1 boost ($+61\%$) does not survive re-seeding, so we frame the effect as KZ retention rather than KY improvement. UZ PPL degrades under E6 (124.19 vs. 56.85), indicating that two-step Kipchak specialization (KZ then KY) displaces UZ-useful representations.

Cross-family control (E6b). Same setup as E6 but warm-started from the English baseline (E8). If any warm-start were sufficient, EN→KY would preserve EN’s cross-lingual profile in the same way. It does not: EN→KY reaches KZ PPL 48.8 ± 1.4 across two seeds—within direct-training spread (50.0 ± 2.4), far above E6’s 23.5. Frobenius growth interpolates between the regimes ($1.13\times \ll 3.10\times \ll 8.36\times$), consistent with the warm-start sitting at a graded distance from a good KY solution. Related-language transfer is qualitatively different from arbitrary checkpoint reuse; the mechanism is better initialization, which we quantify directionally in Section 4.6.

4.6 Directional Geometry of LoRA Solutions

Both spectral energy (Section 4.3) and Frobenius growth (Section 4.4) are functions of the singular-value spectrum of $\Delta W=BA$, and both fail to sepa-

rate $r=64$ collapse (E5) from $r=64$ healthy adaptation (E5c). A natural next probe is *direction*: do the two configurations move the weights along the same axis in weight space, or along incompatible axes? We measure this by per-module Frobenius cosine between every pair of final adapters,

$$\cos_m(\Delta W_X, \Delta W_Y) = \frac{\langle B_X A_X, B_Y A_Y \rangle_F}{\|B_X A_X\|_F \|B_Y A_Y\|_F}, \quad (1)$$

computed without materializing the $\text{out} \times \text{in}$ product (Section S). We report the mean and median of $\cos_m$ over the $42 \times 7 = 294$ LoRA-equipped projections in Gemma-2-9B.

The 22-pair sweep in Table 3 and Figure 1 reveals a sharp bimodal structure that none of the spectral or Frobenius diagnostics surface.

Three regularities follow. **(D1)** Direct LoRA from base is directionally non-canonical: for every configuration trained from raw base (E1b, E3, E5/E5b/E5c, E8), cross-seed cosine sits at 0.01–0.04 with 0/294 modules above 0.5. **(D2)** Transfer initialisation pins the adapter to its source, not to a canonical direction. The headline E6 cross-seed cosine 0.785 measures two stage-2 warm-starts from a *shared* E1-s42 source ($\cos(\text{E6}, \text{source}) = 0.797$ directly). When the source is reseeded ($\cos(\text{E6-from-E1-s42}, \text{E6-from-E1-s123}) = 0.023$ on Gemma, 0.060 on Llama-3) the cosine falls into the direct-training band; transfer propagates source direction rather than producing canonicity. Functional retention (KZ PPL halving), by contrast, reproduces under independent reseeding on both architectures. **(D3)** Collapse and healthy share the same orthogonal sphere: $\cos(\text{E5}, \text{E5b}) = 0.011$, $\cos(\text{E5}, \text{E5c}) = 0.010$, both *inside* the cross-seed band. Direction does not separate the two regimes any more reliably than reshuffling the seed does—a quantitative refutation of H3 and the starting point for the underdetermination argument of Section 5.2, which (D2) extends to transfer once the source is reseeded.

Further directional results. At matched $r/\text{lr}/\text{epochs}/\text{seed}/\text{data}$ the $\alpha=32$ control C5 sits directionally closer to the $r=16$ baseline than to $r=64$ E5c, so α moves the adapter further in ΔW space than rank does; the cosines are robust to the choice of “final” checkpoint (± 0.006) and stable across training; and the Llama-3-8B sweep reproduces every pattern, including the inequality ($0.072 > 0.029$) and source-init pinning (0.804). Section S gives all three in full.

5 Analysis and Discussion

5.1 Why Spectral Energy Fails

Standard LoRA initializes $A \sim \text{Kaiming}$, $B = 0$. After the first gradient update B acquires near-uniform small singular values yielding $\text{SE} \approx 1/r$. Training grows all singular values jointly, distributed across multiple directions, so SE stays stable or decreases. This is rank-biased: $1/64 \approx 0.016$ at $r=64$ vs. $1/16 = 0.063$ at $r=16$, a $3\text{--}4\times$ gap independent of any functional outcome (*higher-rank adapters appear spectrally healthier by definition*). Fundamentally SE is a shape metric invariant to rescaling: spectra $[100, 50, 25]$ and $[10^{-3}, 5 \times 10^{-4}, 2.5 \times 10^{-4}]$ have identical SE despite vastly different perturbation magnitudes.

5.2 Underdetermination: No Final-Adapter Invariant Distinguishes Collapse from Healthy

Section 4.6 supports a stronger negative claim than “SE is the wrong shape metric.” Let $D : (A, B) \mapsto \mathbb{R}$ be any Lipschitz scalar diagnostic (under any reasonable weight-space metric: Frobenius, cosine, or any invariant of $A^\top A$, $B^\top B$, BA). Reliable discrimination of collapse from healthy requires the between-configuration gap Δ_D to exceed the within-configuration spread σ_D^{seed} . The pairwise cosines provide direct evidence at $n=3$ (seeds 42, 123, 7): three cross-seed pairs per configuration, hence a measured spread rather than a single number. Within-config: E5c 0.0316 ± 0.0002 , E5 0.0221 ± 0.0003 , E3 0.0423 ± 0.0013 (mean $\pm$ SD). Between-config, at each seed: E5 vs. E5c 0.0101 (max 0.0103), E5 vs. E3 0.0053. The inequality holds strictly—*smallest* within-config exceeds *largest* between-config—for every pair, by $3\text{--}8\times$, with the within-config spread two orders of magnitude below that margin. So *within the seed-noise regime we measure*, any Lipschitz D has $\text{SNR} \lesssim 1$. This rules out spectral-shape metrics (SE, effective rank, stable rank, entropy), Frobenius scale metrics, and comparisons against any fixed reference—our W_0 -alignment probes fail empirically too. Only *functional* and *trajectory* probes escape the Lipschitz-in- (A, B) assumption. The Llama-3 replication satisfies the same inequality (within-config 0.072 vs. between-config 0.029; Section T), confirming the regime is architecture-independent. The argument *extends to transfer once the source is reseeded*: the architecturally-fair E6 cross-seed is 0.023 (Gemma) and 0.060 (Llama-

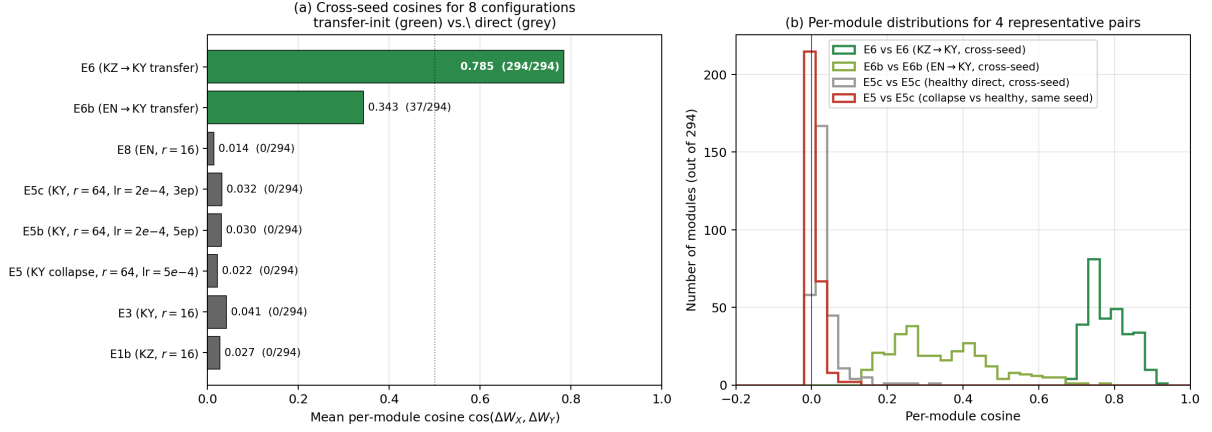


Figure 1: Per-module Frobenius cosine between LoRA $\Delta W=BA$ matrices. **(a)** Cross-seed cosines for the eight seed-replicated configurations under their original protocol. Six direct-from-base configurations (grey: E1b, E3, E5, E5b, E5c, E8) cluster at mean cosine 0.01–0.04 with 0/294 modules above 0.5; the two transfer-initialised configurations (green: E6, E6b) stand apart at 0.785 and 0.343 with 294/294 and 37/294 modules above 0.5. The E6 separation reflects two stage-2 warm-starts from a shared E1-s42 source; the new independent-init test in Table 3 gives 0.023. **(b)** Per-module distributions for four representative pairs. The bimodality is reproduced at the module level: E6 vs. E6 (KZ→KY across seeds) forms a tight peak near 0.8; E6b vs. E6b is broader; both same-configuration direct training (E5c vs. E5c) and the collapse-vs-healthy comparison (E5 vs. E5c) collapse onto a single narrow peak near 0. Direct LoRA from base is directionally non-canonical at the module level, not just on average.

Pair	mean	med.	>0.5
<i>Cross-seed, same config</i>			
E3 vs E3 [†]	0.042	0.026	0/294
E5 vs E5 [†]	0.022	0.015	0/294
E5b vs E5b	0.030	0.018	0/294
E5c vs E5c [†]	0.032	0.019	0/294
E8 vs E8	0.014	0.009	0/294
E1b vs E1b	0.028	0.016	0/294
E6 vs E6 (KZ→KY, shared E1-s42 init)	0.785	0.774	294/294
E6 vs E6 (KZ→KY, independent E1-s42/E1-s123 inits)	0.023	0.011	0/294
E6b vs E6b (EN→KY)	0.343	0.324	37/294
<i>Collapse vs healthy at $r=64$</i>			
E5 vs E5b	0.011	0.006	0/294
E5 vs E5c	0.010	0.006	0/294
E5b vs E5c	0.053	0.033	0/294
<i>α vs. rank (C5)</i>			
E5c vs C5 ($\alpha=128/32$)	0.072	0.053	0/294
C5 vs E3 ($r=64/16$)	0.155	0.139	2/294
C5 vs E5	0.009	0.005	0/294

Table 3: Per-module Frobenius cosine between LoRA $\Delta W=BA$ matrices, aggregated over the 294 attention/MLP projections in Gemma-2-9B. The E6 *shared-init* row reproduces the original headline (0.785, all 294 modules > 0.5): two stage-2 warm-starts from the same E1-s42 source converge to a near-identical direction. The E6 *independent-init* row (new in this revision: E6-s42-from-E1-s42 vs. E6-indep-s123-from-E1-s123, both stages independently re-seeded) gives 0.023 – inside the direct-training band. **Transfer pins to its source, not to a canonical direction.** The Llama-3 architecturally-fair test gives 0.060 (Section T). Crucially, $\cos(E5, E5c) = 0.010$ ($r=64$ collapse vs. healthy) is *indistinguishable* from same-configuration cross-seed pairs (0.022, 0.032): the two basins are no further apart than two seeds of the same setup. The C5 control ($r=64$, $\alpha=32$) is directionally closer to E3 ($\cos 0.155$, different rank, same α) than to E5c ($\cos 0.072$, same rank, different α)— α moves the adapter further in ΔW space than rank does. [†]Replicated at $n=3$ (seeds 42, 123, 7); the value shown is the mean over the three cross-seed pairs, whose spread is ± 0.0013 (E3), ± 0.0003 (E5), ± 0.0002 (E5c). The collapse-vs-healthy gap (0.0103 at its largest across seeds) therefore sits an order of magnitude below the within-configuration band, not merely below a single measurement of it. Cross-language, transfer-vs-direct, and other-controls pairs in Table 17.

3), inside the direct-training band—transfer pins the trajectory to its source, not to a canonical direction.

What generalises, and how far. Our four claims carry different scope. (1) *The SE-threshold falsification is existential* and travels furthest: falsifying a diagnostic needs reproduced counterexamples, not broad coverage, and we give four sub-threshold $r=64$ runs spanning the full functional range, with the collapse case at three seeds on Gemma and reproduced on Llama-3. We do not claim spectral quantities are uninformative in general—only that this threshold is not a reliable gate here. (2) *The underdetermination inequality is a measured regime property* of direct LoRA on two architectures; it bounds the SNR of Lipschitz functionals *in the seed-noise regime we measure*, and settings with more canonical solutions (much larger data, different initialisation, or shared-source transfer) need not satisfy it. (3) *Frobenius growth* and (4) *related-language retention are comparative claims* resting on three Turkic languages plus English: retention reproduces across architectures and under independent reseeding, but three languages cannot establish a general law. The recommendations below follow from (1) and (2), the claims we consider portable.

Recommendations. (i) Do not gate fine-tuning health on SE thresholds; (ii) track $\|B\|_F$ trajectories paired with cross-lingual PPL probes (not fixed numeric cutoffs); (iii) treat collapse detection as a functional test (no Lipschitz $D(A, B)$ can substitute); (iv) use related-language transfer initialisation when available—it robustly preserves source-language knowledge (functional retention is the strong claim), with the caveat that directional reproducibility across seeds depends on a shared source. Connections to mode-connectivity (Garipov et al., 2018; Frankle et al., 2020), intrinsic-dimensionality (Aghajanyan et al., 2021), LP-FT (Kumar et al., 2022), and PiSSA / LoRA+ (Meng et al., 2024; Hayou et al., 2024) are discussed in Section P.

6 Conclusion

We presented a systematic study of LoRA spectral and directional dynamics for three low-resource Turkic languages, replicated across two architectures. (1) The spectral-energy threshold is not predictive: sub-threshold runs span the full range from catastrophic collapse to healthy baseline. The collapse is driven by learning rate, the ap-

parent benefit of higher rank is an artefact of the $\alpha=2r$ convention, and a full-precision control rules out quantization. (2) Frobenius growth separates regimes within a language category but fails across categories. (3) Structural underdetermination: cross-seed directional variance is commensurate with the collapse-vs-healthy gap, so within the measured regime no Lipschitz function of the final adapter alone can separate the two; detection must come from functional or trajectory probes. (4) Related-language warm-start robustly preserves source-language knowledge on both architectures, while a cross-family warm-start does not; directionally, transfer pins the adapter to its source, and transfers from independent sources land orthogonally—extending the underdetermination argument to transfer. Future work should test PiSSA-style (Meng et al., 2024) initialisations and trajectory diagnostics that escape the underdetermination argument.

Limitations

Our study has several important limitations, which we discuss here in decreasing order of severity for downstream interpretation.

(L1) Limited-seed experiments. The three configurations carrying the underdetermination argument (E3, E5, E5c) are replicated at $n=3$ (seeds 42, 123, 7), so the inequality of Section 5.2 rests on a measured within-config spread rather than a single cross-seed number. The remaining Gemma-2 configurations are at $n=2$ (only C3 BF16 single-seed), as are all five Llama-3 configurations—L3-C5 seed-replicated in this revision; see Section J. That C5 replication also shows the inequality is not universal across configuration *pairs*: on Llama-3, $\cos(C5-s42, C5-s123) = 0.200$ falls below $\cos(C5, E3) = 0.343$ (Section T). It holds for the collapse-vs-healthy comparisons the argument rests on, on both architectures; we do not claim more. The data confirm six claims across both seeds: $2 \times$ KZ-PPL retention of related-language transfer (50.0 ± 2.4 direct vs. 23.5 ± 0.1 E6); cross-family transfer does *not* reproduce it (E6b 48.8 ± 1.4); E5c $>$ E3 (3.89 ± 0.03 vs. 4.81 ± 0.03); E5b stays outside E5’s collapsed regime; E1b data-vs-coverage (4.16 ± 0.01); and EN-vs-Turkic Frobenius dissociation (E8 $33.3 \pm 0.1 \times$ no forgetting vs. KZ baseline $38.5 \times$ severe forgetting). Extending $n=3$ to the transfer configurations and adding a second BF16 seed would tighten the remaining quantita-

tive comparisons; the two-seed results are reported as consistent-across-seeds rather than as variance estimates.

(L2) Training-set contamination for NER (quantified). Our Turkic training corpora include Wikipedia dumps, and the WikiANN evaluation set is itself derived from Wikipedia entity links (Pan et al., 2017). To quantify this overlap, we computed the fraction of unique gold WikiANN-test entity strings that appear verbatim somewhere in each training corpus (case-insensitive substring match): **Kazakh 65.7% (497/757), Kyrgyz 55.9% (52/93), Uzbek 2.2% (13/602)**. The Kazakh and Kyrgyz NER F1 numbers in Table 2 therefore upper-bound the contribution of memorization at $\sim 60\%$, and cross-configuration NER comparisons within those two languages are partially confounded by this shared-entity floor. The Uzbek overlap of 2%, by contrast, is below the resolution of memorization (this turns out to be a script-mismatch artifact—see L9—which makes Uzbek a clean cross-lingual probe). The transfer-induced UZ improvements in Table 2 (e.g., E1’s 0.578 F1 on UZ) are thus most likely *not* memorization. A non-Wikipedia NER benchmark such as KazNERD (Yeshpanov et al., 2022) (gated; access pending) would tighten this further; the per-type breakdown is reported in Section H.

(L3) Rank–LR–epoch confound (now closed by E5b, E5c, and C5). Experiment E5 ($r=64$, $\text{lr}=5 \times 10^{-4}$, 5 epochs, $\alpha=128$) originally varied four factors. E5b matches rank but lowers LR; E5c additionally matches epoch count; C5 additionally matches α . Together they decompose the variation cleanly (Section 4.1). Catastrophic collapse = aggressive base LR (refuted by E5b). Extra cross-lingual forgetting = training duration (refuted by E5c). *Apparent* rank-only target-language advantage = amplification of absolute α by the $\alpha=2r$ convention (which at $r=64$ sets $\alpha=128$, $4 \times$ E3’s $\alpha=32$); refuted by C5: holding absolute $\alpha=32$ at $r=64$, the adapter is indistinguishable from the $r=16$ baseline on KY/KZ/UZ PPL, NER F1, and TUMLU. Rank itself, at matched absolute α , does not deliver a functional benefit in our setting.

(L4) Quantization confound (addressed by C3). The C3 BF16 control (Kyrgyz, $r=16$, identical to E3 except for full-precision base) refutes the concern that 4-bit NF4 quantization suppresses singular-value growth: BF16 SE is statistically indistinguishable from 4-bit E3 (final max SE 0.32 vs. 0.33 ± 0.01 , peak 0.47 vs. 0.45 ± 0.02 ; Table 11).

Frobenius, KY PPL, NER, TUMLU are within seed variance of E3 (Section I). C3 is $n=1$ only; a second BF16 seed would tighten this.

(L5) Corpus imbalance (closed by E1b, E2b, KZ-4.4M). Byte-matched corpora differ post-tokenization (KZ 14.1M, UZ 5.4M, KY 4.4M tokens). Three subsampled controls decompose volume from tokenizer/pretraining coverage. KZ: E1b ($\sim 1.5\text{M}$) gives PPL 4.17 vs. E1’s 2.73 at full corpus; the new true token-matched KZ-4.4M control gives 3.61, still below KY baseline’s 4.78 at the same budget – coverage advantage confirmed. UZ: the new E2b ($\sim 1.5\text{M}$) gives 7.01 vs. E2’s 6.18 – relative volume penalty much smaller on UZ. Both factors contribute; neither alone explains the cross-language gaps.

(L6) NER as a primary metric. Our NER evaluation uses 3-shot prompting with $n=100$ samples per language and F1 derived from parse-success output. This is closer to a test of instruction-following than of entity knowledge. The $r=64$ “NER F1 = 0” result, in particular, is driven by 100% parse failures; the model may retain some entity knowledge that this evaluation cannot access. Log-likelihood-based NER scoring (scoring candidate entity spans by model probability) would separate instruction loss from knowledge loss and is a priority follow-up.

(L7) Scope of generalization. All four paper-level findings replicate on Llama-3-8B across five configurations including the C5 absolute- α control and a two-stage E6 KZ→KY transfer (Section T); Llama-3 cosines run $\sim 2 \times$ Gemma’s but the regime structure is identical. We test only three of ~ 35 Turkic languages at fixed ~ 150 MB/language; SVD monitoring uses `lora_B`, pairwise cosines use full $\Delta W=BA$.

(L8/L9) Two further limitations in Section M: (L8) Section 4.4 growth ranges are within-sample, not held-out thresholds; (L9) Cyrillic-only UZ corpus vs. Latin WikiANN-uz explains L2’s 2% overlap and bounds UZ generalisability.

Ethical considerations

This work involves fine-tuning language models on text corpora from three Turkic languages and English. While we aim to improve NLP capabilities for underserved language communities, we acknowledge that language models can amplify biases present in training data. Our training corpora, drawn from Wikipedia, literary texts, and

web sources, may contain cultural biases or factual inaccuracies.

Data availability: The Kazakh and Uzbek corpora are compiled from publicly available sources (Wikipedia, open book collections). The Kyrgyz corpus is curated from licensed literary and encyclopedic sources and cannot be redistributed due to copyright restrictions. In the spirit of open science, we release all experimental code, SVD monitoring logs, and evaluation results via an anonymous repository (see Reproducibility Statement). Trained LoRA adapters will be publicly released upon acceptance.

Reproducibility Statement

All experimental code (training with SVD monitoring, evaluation, and plotting scripts), SVD monitoring logs, and evaluation reports are publicly available at an anonymous repository.⁴ Fine-tuned LoRA adapters for all Gemma-2-9B and Llama-3-8B configurations will be released on a public model hub upon acceptance. The Kazakh, Uzbek, and English training corpora are derived from publicly available sources and can be reconstructed using the provided data preparation scripts. The Kyrgyz corpus cannot be redistributed due to licensing restrictions on its source materials; consequently, the KY training stage is not fully reproducible. However, the released KY adapters and SVD logs enable full reproduction of all evaluation and analysis stages.

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A Additional Figures

B Cross-Lingual PPL Matrix (full)

Trained on	KY	KZ	UZ
KZ Baseline (14.1M tok, E1)	40.75	2.73	41.27
KZ Small-corpus (E1b)	25.55	4.17	23.17
UZ Baseline (E2)	116.89	64.44	4.03
KY Baseline (E3)	4.78	47.58	56.85
KY Overfit (E4)	4.18	87.65	95.76
KY $r=64$ (E5)	5.90	659.25	742.73
KY $r=64$ (E5b)	4.55	128.04	162.68
KY $r=64$ (E5c)	3.86	93.52	111.25
KZ→KY (E6)	4.73	23.49	124.19
EN→KY (E6b)	4.78	47.46	63.42
EN Control (E8)	14.96	5.54	16.55
KY BF16 (C3)	4.99	43.97	59.02
KY $r=64$ $\alpha=32$ (C5)	4.46	49.22	57.36

Table 4: Cross-lingual perplexity. Bold = target language. Reported on the same KY/KZ/UZ held-out splits across all adapters.

C Data Volume vs. Pretraining Coverage (E1 vs. E1b)

To disentangle whether E1’s strong target PPL (2.73) reflects $3\times$ larger token count or better tokenizer/pretraining coverage for Kazakh, the small-corpus control E1b runs on ~ 1.5 M training tokens. Target KZ PPL rises to 4.17, showing data volume contributes ~ 1.4 PPL of the E1 advantage. Critically, E1b’s 4.17 remains below the KY baseline’s 4.78 at $\sim 3\times$ fewer tokens, so Kazakh enjoys a genuine pretraining-coverage advantage independent of corpus size. Cross-lingually E1b also shows lower forgetting than E1 (KY 25.55 vs. 40.75; UZ 23.17 vs. 41.27), consistent with fewer steps inducing milder perturbations.

D Frobenius Norm Growth (full table)

E Detailed Training Configurations

All experiments were conducted on a single consumer GPU (16 GB VRAM) using PyTorch 2.10, Transformers 5.2, and PEFT 0.18. Table 7 provides the full training statistics.

SVD Dynamics: Baseline vs Aggressive vs Transfer

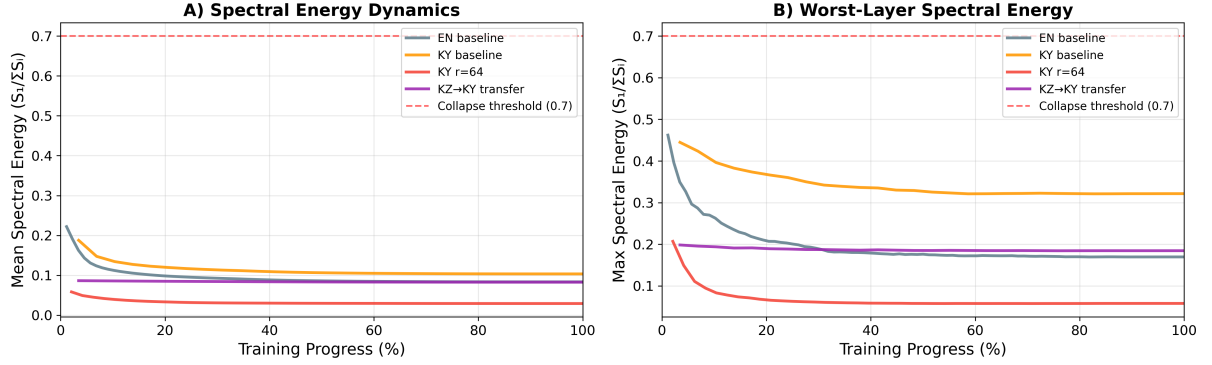


Figure 2: Spectral energy dynamics during training. Top: mean SE; bottom: max SE. No experiment approaches the proposed 0.7 threshold; the $r=64$ row even shows *decreasing* SE while suffering the worst functional degradation.

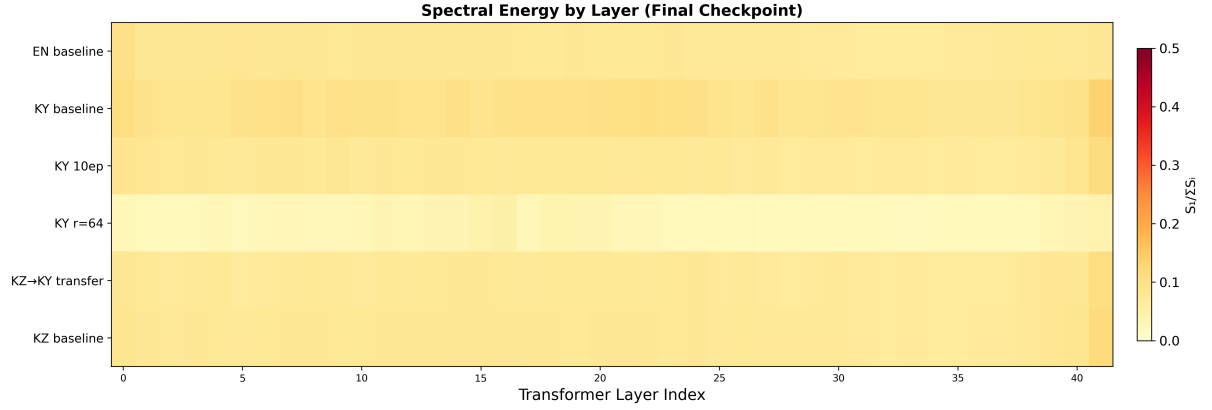


Figure 3: Layer-wise spectral energy at the final training checkpoint. Each row is an experiment; each column is a transformer layer. No layer in any experiment exceeds 0.7.

Frobenius Norm Dynamics: The Missing Diagnostic

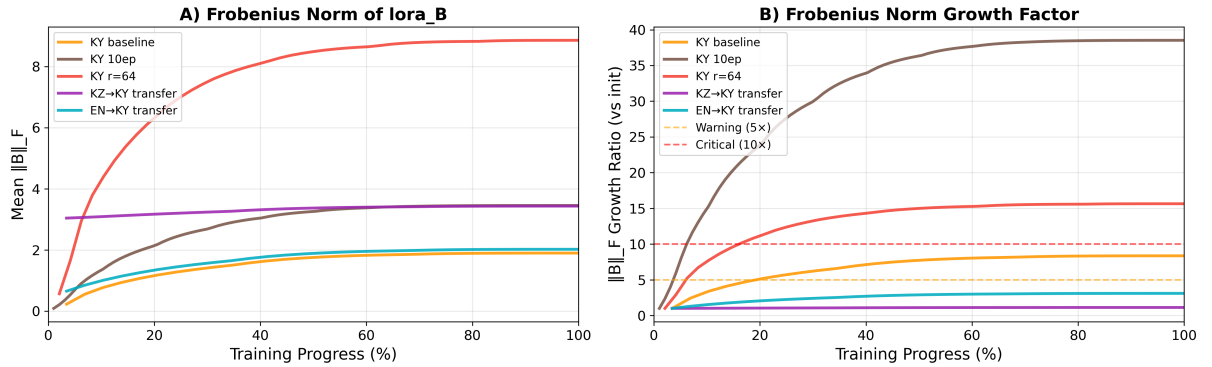


Figure 4: Frobenius norm dynamics of `lora_B` matrices during training. $r=64$ shows dramatically larger growth than $r=16$; KZ→KY transfer begins at the KZ baseline's final norm and only adds +13%.

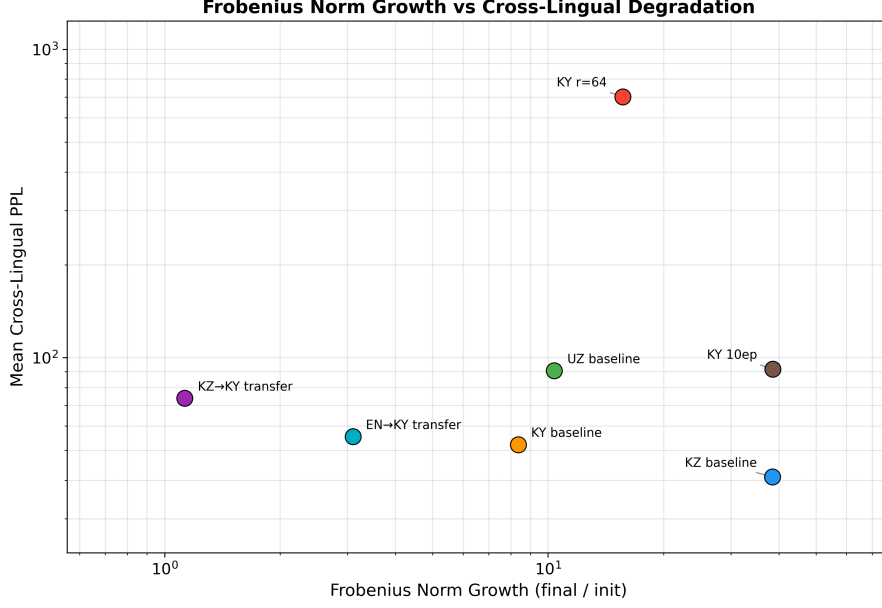


Figure 5: Frobenius norm growth vs. cross-lingual PPL degradation. Within Turkic the trend is monotone (more growth, more forgetting), but the EN control sits at $33\times$ growth with minimal forgetting—norm growth alone is not sufficient to predict catastrophic forgetting.

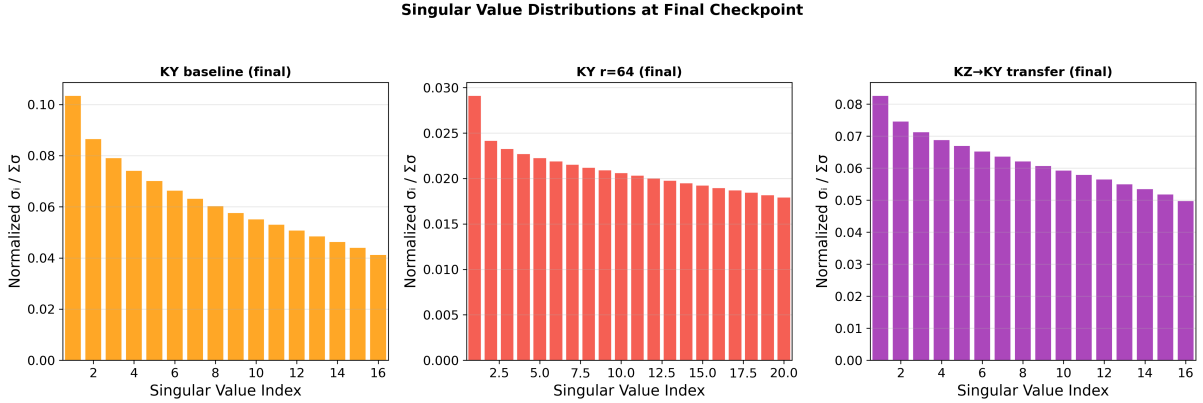


Figure 6: Singular value distributions at the final checkpoint. The $r=64$ configuration develops a well-distributed spectrum (low SE) with dramatically larger absolute magnitudes than the $r=16$ baseline—explaining why SE fails to detect pathological training.

Experiment	$\ \mathbf{B}\ _F^{\text{init}}$	$\ \mathbf{B}\ _F^{\text{final}}$	Growth
KZ→KY (E6, related)	3.043	3.436	$1.13\times$
EN→KY (E6b, cross-family)	0.653	2.023	$3.10\times$
KY Baseline $r=16$ (E3)	0.227	1.899	$8.36\times$
EN Control $r=16$ (E8)	0.092	3.076	$33.4\times$
KY BF16 $r=16$ (C3)	0.218	1.880	$8.6\times$
KZ Baseline $r=16$ (E1)	0.081	3.107	$38.5\times$
KY $r=64$ lr 5×10^{-4} 5ep (E5)	0.566	8.859	$15.6\times$
KY $r=64$ lr 2×10^{-4} 5ep (E5b)	0.247	4.486	$18.2\times$
KY $r=64$ lr 2×10^{-4} 3ep (E5c)	0.388	3.385	$8.7\times$
KY $r=64$ $\alpha=32$ 3ep (C5)	0.456	3.900	$8.55\times$
Diverged $r=64$ (E7)	—	—	explodes

Table 5: Frobenius norm growth ratios for `lora_B`. “Init” is the first SVD-log measurement (training step 100), not the literal loaded checkpoint. For E6 the step-100 norm stays close to the loaded KZ-final norm (3.04 vs. 3.11); for E6b the step-100 norm drops sharply to 0.65 from EN-final 3.08, consistent with EN-init directions being misaligned with KY gradients and partially zeroed before the adapter rebuilds.

ID	Experiment	Language	Rank	LR	Epochs	Trainable %	Init
E1	KZ Baseline (14.1M tok)	Kazakh	16	2×10^{-4}	3	1.05%	Random
E1b	KZ Small-corpus (1.5M tok)	Kazakh	16	2×10^{-4}	3	1.05%	Random
E2	UZ Baseline	Uzbek	16	2×10^{-4}	3	1.05%	Random
E3	KY Baseline	Kyrgyz	16	2×10^{-4}	3	1.05%	Random
E4	KY Overfit	Kyrgyz	16	2×10^{-4}	10	1.05%	Random
E5	KY $r=64$ high-LR	Kyrgyz	64	5×10^{-4}	5	4.08%	Random
E5b	KY $r=64$ std-LR	Kyrgyz	64	2×10^{-4}	5	4.08%	Random
E5c	KY $r=64$ epoch-matched	Kyrgyz	64	2×10^{-4}	3	4.08%	Random
E6	KZ→KY Transfer	Kyrgyz	16	2×10^{-4}	3	1.05%	KZ checkpoint
E6b	EN→KY Transfer [◊]	Kyrgyz	16	2×10^{-4}	3	1.05%	EN checkpoint
E7	Diverged	KY [‡]	64	1×10^{-3}	3	4.08%	Random
E8	EN Control	English	16	2×10^{-4}	3	1.05%	Random
C3	KY BF16* control	Kyrgyz	16	2×10^{-4}	3	0.58%	Random
C5	KY $r=64$ $\alpha=32$ ^b	Kyrgyz	64	2×10^{-4}	3	4.08%	Random

Table 6: Summary of experimental configurations. All Gemma-2-9B experiments except C3 use 4-bit NF4 quantization. C3 uses full bfloat16 precision and serves as the quantization control. All runs except C5 use $\alpha = 2r$; C5 uses $\alpha=32$ at $r=64$ as the matched-absolute- α control for the rank-only claim of E5c (same absolute α as E3, breaking the $\alpha=2r$ convention which would set $\alpha=128$). [‡]E7 diverged at step 800; excluded from downstream eval. *C3 trainable% differs because PEFT counts BF16 vs. 4-bit base parameters differently. [◊]E6b warm-starts from E8 instead of E1. ^bC5 is otherwise identical to E5c.

ID	Experiment	Train Loss	Eval Loss	Eval PPL	Tokens	Time (min)
E1	KZ Baseline (14.1M)	0.275	1.259	3.52	14.1M	477
E1b	KZ Small-corpus (1.5M)	1.186	1.461	4.31	1.5M	122
E2	UZ Baseline	1.193	1.822	6.18	5.4M	341
E3	KY Baseline	0.815	1.920	6.82	4.4M	183
E4	KY Overfit	1.050	1.963	7.12	4.4M	1193
E5	KY $r=64$ high-LR	0.274	2.164	8.71	4.4M	157
E5b	KY $r=64$ std-LR (5ep)	1.278	1.964	7.13	4.4M	605
E5c	KY $r=64$ epoch-matched	1.652	1.898	6.67	4.4M	360
E6	KZ→KY	1.809	1.932	6.90	4.4M	355
E8	EN Control	1.821	2.017	7.51	12.7M	1351

Table 7: Full training statistics for all experiments. Eval loss and PPL are computed on the validation split during training (distinct from the post-hoc evaluation in Table 2). The higher train loss for E6 (KZ→KY) reflects the adapter starting from KZ-optimized weights that initially perform poorly on KY data. **Wall-time variance:** E5 vs. E5b (157 vs. 605 min) and E5c vs. E3 (360 vs. 183 min) reflect GPU-contention on a shared consumer GPU, not algorithmic differences; `total_flos` is identical between E5/E5b/E5c (1.02×10^{18}), and `samples_per_second` reported in the run logs varies between ~ 2 and ~ 8 across these runs solely due to scheduling.

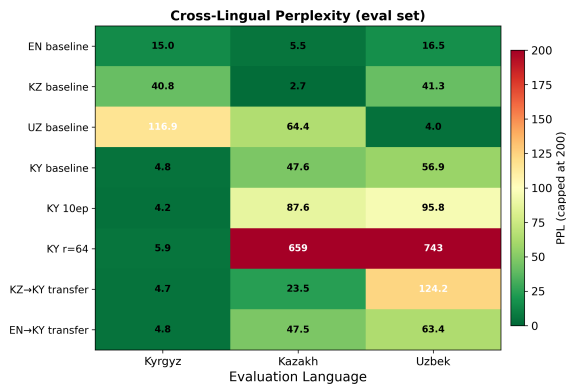


Figure 7: Cross-lingual PPL heatmap (color clipped at 200). The $r=64$ high-LR row is pathological; KZ→KY shows the most balanced profile.

F Effective Rank Dynamics

Figure 8 shows the effective rank dynamics across experiments. The $r=64$ configuration shows increasing effective rank (from ~ 6 to ~ 26), indicating growing parameter utilization. The diverged experiment (E7) is the only configuration where effective rank *decreases* (from 43.3 to 37.1), consistent with true spectral collapse.

E7 divergence analysis. The diverged experiment ($r=64$, $\text{lr}=10^{-3}$) provides the most extreme datapoint. Training terminated at step 800 due to loss divergence. Even at this point, maximum SE was only 0.18—still far below the 0.7 threshold—while mean Frobenius norm had already reached 6.75 ($19\times$ growth in just 800 steps, compared to

15.6 $\times$ over 4,885 steps for E5). The effective rank decrease (43.3 $\rightarrow$ 37.1) in the final 200 steps is the only instance of true rank contraction in our experiments, suggesting that SE-based collapse detection may only trigger *after* the model has already diverged—too late for practical intervention. The Frobenius norm, by contrast, shows clear exponential growth from step 100 onward, providing a much earlier warning signal.

G Log-Likelihood NER: Type Classification under Gold Spans

To separate loss of entity knowledge from loss of instruction-following/output-formatting, we add a second NER evaluation that does not require the model to generate structured output. Given the gold-span boundaries from WikiANN, we present each span in context and score three candidate completions (“person”, “organization”, “location”) by log-likelihood under the adapter; the predicted type is the argmax. Because this evaluation never relies on the model’s output format, it is immune to the 100% parse-failure mode that drives E5’s NER F1 = 0 under generation-based scoring.

Table 8 reports per-language type accuracy and macro-F1 on the same 100-example test sets used for Table 2.

H WikiANN–Training Entity Overlap

We quantify the entity-level overlap between each language’s training corpus and the WikiANN test split it is evaluated on. For every gold span in the WikiANN test set we test whether the entity string appears verbatim (case-insensitive) anywhere in the corresponding training corpus. The unique-overlap rate is an upper bound on the fraction of NER F1 that could be explained by memorization of seen entity surface forms.

I BF16 vs. 4-bit Control (C3)

Table 11 reports a side-by-side comparison of the 4-bit E3 baseline (averaged across our two seeds where available) and the full-precision BF16 control (C3, single seed, run on a cloud A100 40GB instance). All hyperparameters except quantization are identical.

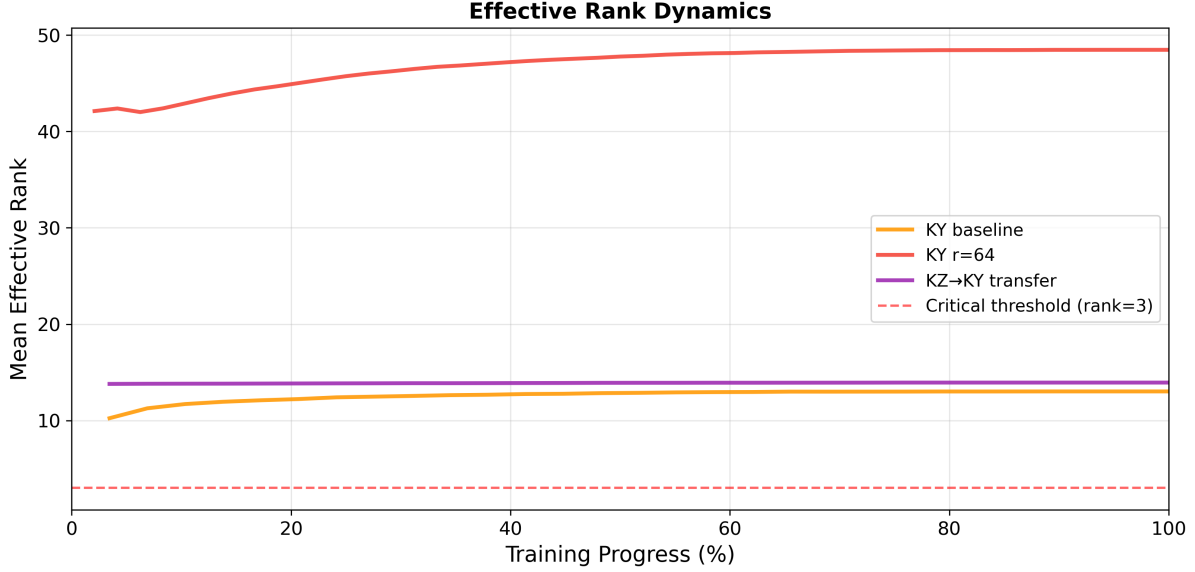


Figure 8: Effective rank dynamics during training. The $r=64$ configuration shows increasing effective rank despite functional degradation, further evidence that standard spectral metrics can be misleading.

Experiment	KY	KZ	UZ
E1 KZ Baseline (14.1M tok)	57.7%	72.3%	70.9%
E1b KZ Small-corpus (1.5M)	46.0%	70.5%	69.9%
E2 UZ Baseline	54.9%	71.4%	65.0%
E3 KY Baseline	50.4%	69.6%	63.1%
E4 KY Overfit (10ep)	56.8%	71.4%	60.2%
E5 KY $r=64$, lr 5×10^{-4}	59.5%	50.0%	55.3%
E5b KY $r=64$, 5ep, lr 2×10^{-4}	53.1%	70.5%	69.9%
E5c KY $r=64$, 3ep, lr 2×10^{-4}	54.9%	67.9%	63.1%
E6 KZ→KY (related)	59.5%	73.2%	66.0%
E6b EN→KY (cross-fam.)	54.0%	70.5%	70.9%
E8 EN Control	56.8%	73.2%	68.0%

Table 8: Log-likelihood span-type accuracy across experiments (gold-span boundaries, argmax over {PER, ORG, LOC}). Random-baseline accuracy is 33.3%. The parse-failure-immune evaluation reveals that **E5 retains and even marginally exceeds target-language (KY) entity knowledge** (59.5%, tied for best with E6) despite scoring F1 = 0 under generation-based NER—confirming that the high-LR $r=64$ collapse destroys instruction-following, not entity knowledge itself. Cross-lingual degradation, however, does appear in E5 on non-target languages (KZ 50.0% vs. 70–73% for others), consistent with the cross-lingual PPL increase. Under this cleaner metric, the KZ→KY transfer advantage over the KY baseline is modest (59.5% vs. 50.4% on KY; +9.1 absolute percentage points) rather than the +61% relative improvement suggested by generation F1, reinforcing our framing of cross-lingual transfer as a *directional* rather than precise effect. Notably, the cross-family transfer E6b (EN→KY) sits intermediate at 54.0%, consistent with cross-family initialization providing partial—but smaller—entity-knowledge benefit than related-language transfer.

Language	Total spans	Unique	Unique overlap
Kazakh (KZ)	1,115	757	65.7% (497/757)
Kyrgyz (KY)	111	93	55.9% (52/93)
Uzbek (UZ)	1,011	602	2.2% (13/602)

Table 9: Verbatim case-insensitive overlap between WikiANN test entities and our training corpora. KZ and KY show high overlap (memorization upper bound $\sim 60\%$); UZ shows almost none, which we attribute to script mismatch between our Cyrillic UZ corpus and Latin-script WikiANN-uz (L9). The UZ low overlap incidentally turns Uzbek into a cleaner cross-lingual transfer probe: improvements in UZ NER are unlikely to be memorization artefacts.

Language	PER	ORG	LOC
Kazakh	71% (221/313)	62% (170/276)	63% (106/168)
Kyrgyz	33% (11/33)	57% (17/30)	80% (24/30)
Uzbek	3% (2/67)	4% (8/202)	1% (3/333)

Table 10: Per-type unique overlap. The KY pattern (LOC $\gg$ PER) is consistent with location names being a comparatively closed-class compared to person names. UZ remains uniformly low across types, confirming the script-mismatch explanation.

Metric	E3 (4-bit, 2 seeds)	C3 (BF16, $n=1$)
KY PPL	4.81 ± 0.02	4.99
KZ PPL (cross-ling)	50.0 ± 2.4	43.97
UZ PPL (cross-ling)	55.5 ± 1.4	59.02
NER F1 KY (gen)	0.170 ± 0.013	0.164
NER TypeAcc KY	55.0 ± 4.6	58.6
TUMLU KY	36.1 ± 1.4	36.1
$\ B\ _F$ growth	$8.36\times$	$8.60\times$
Final SE max	0.33 ± 0.01	0.32
Final SE mean	0.10	0.10
Peak SE max (training)	0.45 ± 0.02	0.47

Table 11: 4-bit NF4 vs. BF16 control on the same KY baseline configuration (rank 16, lr 2×10^{-4} , 3 epochs). E3 (4-bit) values are reported as mean $\pm$ half-range across two seeds (seeds 42 and 123); we use half-range rather than standard deviation because the unbiased SD estimator at $n=2$ is unreliable, and half-range gives a more honest visual sense of seed-to-seed spread. All downstream metrics fall within or near this two-seed spread for the 4-bit run. Frobenius norm growth is essentially identical. Spectral energy in BF16 is statistically indistinguishable from 4-bit at every comparable point (final max, final mean, in-training peak), which rules out the original concern: 4-bit quantization is not artificially suppressing SE. The SE-threshold falsification therefore carries over to full precision. We did not run a second seed for C3 due to compute cost (the BF16 run was completed on rented cloud GPU); $n=1$ is acknowledged in L1.

J Seed-Variance Replication

To provide at least partial seed-variance estimates for our most consequential configurations, we rerun two experiments under a second seed (123, while the originals used seed 42). Training hyperparameters and data are identical across seeds; only the random seed for initialization, data shuffling, and train/val split differs.

Three additional cross-seed pairs (Table 13) extend the seed-replication to the two remaining Gemma baselines (E1 KZ, E2 UZ) and the Llama-3 collapse run (L3-E5). All three show the same behaviour-vs-direction dissociation that anchors the underdetermination argument: target-language PPL is tight across seeds (within 0.04–0.12), yet the per-module weight-space direction is near-orthogonal (cross-seed cosine 0.026–0.047, 0 modules above 0.5 on either architecture). The L3-E5 pair additionally confirms that catastrophic functional collapse (NER F1 = 0, TUMLU $\sim$ 25%, KZ PPL > 140) reproduces at both seeds.

These results sharpen the interpretive stance taken throughout the paper: we treat cross-lingual PPL differences as the most reliable quantitative

signal, and the various NER scores as directional indicators whose per-experiment magnitudes should not be over-interpreted. The present two-seed data now covers eleven of twelve Gemma-2 configurations and four of five Llama-3 configurations; a full $n=3$ replication remains the most important reproducibility follow-up (see L1).

K Per-Type NER Results

Table 14 breaks down NER performance by entity type for the Kyrgyz evaluation.

L TUMLU Accuracy by Language

Table 15 presents TUMLU accuracy scores across all languages and experiments.

M Additional Limitations (L8, L9)

(L8) Post-hoc Frobenius observations. The growth ranges in Section 4.4 are derived from the same experiments on which their interpretation is built; they are observations requiring held-out validation, not established thresholds, and we deliberately reframe them as such in Section 4.4.

	E1b		E3		E5		E5b		E5c		E6		E6b		E8 (EN)	
Metric	s=42	s=123	s=42	s=123	s=42	s=123	s=42	s=123	s=42	s=123	s=42	s=123	s=42	s=123	s=42	s=123
KY PPL ↓	25.55	28.81	4.78	4.83	5.90	6.01	4.55	4.62	3.86	3.92	4.73	4.78	4.78	4.83	14.96	14.82
KZ PPL ↓	4.17	4.15	47.58	52.47	659.25	606.81	128.04	107.23	93.52	98.62	23.49	23.61	47.46	50.21	5.54	5.50
UZ PPL ↓	23.17	21.93	56.85	54.05	742.73	763.79	162.68	139.89	111.25	96.61	124.19	114.54	63.42	57.51	16.55	16.28
NER F1 KY (gen) ↑	0.150	0.139	0.157	0.182	0.000	0.000	0.146	0.116	0.115	0.169	0.253	0.179	0.133	0.110	0.209	0.227
TypeAcc KY ↑	46.0	53.1	50.4	59.5	59.5	53.1	53.1	64.9	54.9	64.9	59.5	64.9	54.0	57.7	56.8	54.9
TypeAcc KZ ↑	70.5	70.5	69.6	70.5	50.0	40.2	70.5	71.4	67.9	69.6	73.2	75.0	70.5	69.6	73.2	72.3
TUMLU KY ↑	37.3	38.6	34.7	37.4	23.2	23.9	33.2	33.2	38.5	38.0	33.7	34.9	36.1	35.9	37.7	36.2
$\ B\ _F$ growth	2.86×	2.88×	8.36×	8.30×	15.65×	15.50×	18.19×	18.07×	8.72×	8.66×	1.13×	1.13×	3.10×	3.09×	33.39×	33.17×

Table 12: Two-seed replication for the eight configurations underlying our main claims: small-corpus KZ baseline (E1b, 1.5M tokens), KY baseline (E3), catastrophic collapse (E5), rank-LR control (E5b), epoch-matched control (E5c), related-language transfer (E6: KZ→KY), cross-family transfer (E6b: EN→KY), and EN control (E8). All TypeAcc values are percentages. Aggregate values reported elsewhere as “ $x \pm y$ ” use the seed mean and half-range ($y = (\max - \min)/2$), not standard deviation, because the unbiased SD estimator at $n=2$ is unreliable; readers should interpret these as consistent-across-two-seeds spreads rather than population-variance estimates. **Six claims hold consistently across both seeds:** (i) *related-language transfer halves cross-lingual KZ PPL* (E6: 23.5 ± 0.1 vs. E3 50.0 ± 2.4 , a separation $\sim 10\times$ larger than either condition’s spread); (ii) *cross-family warm-start does not reproduce this retention* (E6b: KZ PPL 48.8 ± 1.4 , within seed spread of direct training, far above E6); (iii) *$r=64$ outperforms $r=16$ baseline at matched conditions* (E5c KY PPL 3.89 ± 0.03 vs. E3 4.81 ± 0.03); (iv) *rank-LR disentanglement* (E5b sustains $F1 \geq 0.12$, TUMLU $\sim 33\%$ in both seeds, far outside E5’s $F1 = 0$, TUMLU $\sim 23\%$); (v) *E1b data-vs-coverage finding* (E1b’s KZ PPL 4.16 ± 0.01 is below E3’s 4.81 despite $\sim 3\times$ fewer tokens, while well above E1’s 2.73 at full corpus); (vi) *EN-vs-Turkic Frobenius dissociation* (E8 $33.3 \pm 0.1 \times$ growth with KZ PPL 5.52 ± 0.02 across seeds, indistinguishable from KZ baseline’s $38.5\times$ with severe forgetting). The single-seed transfer NER F1 headline (+61%) does *not* survive reseeding (-2% at seed 123); we therefore reframe transfer as a retention mechanism.

Config (target)	PPL s42	PPL s123	F1 s42	F1 s123	cross-seed cos (>0.5)
E1 (KZ baseline)	2.73	2.69	0.164	0.214	0.026 (0/294)
E2 (UZ baseline)	4.03	3.91	0.287	0.265	0.047 (0/294)
L3-E5 (collapse)	3.43	3.49	0.000	0.000	0.037 (0/224)

Table 13: Three additional cross-seed pairs. **PPL** is the target-language perplexity (KZ for E1, UZ for E2, KY for L3-E5); **F1** is target-language generation NER F1. Target PPL is tight across seeds for all three, while the cross-seed per-module $\Delta W=BA$ cosine stays in the direct-training band (≤ 0.047 , 0 modules above 0.5) – the same behaviour-vs-direction dissociation as Table 12, now on two more Gemma baselines and on Llama-3. L3-E5 reproduces catastrophic collapse ($F1=0$) at both seeds.

(L9) Uzbek script coverage. Our Uzbek training corpus is restricted to Cyrillic-script sources, while WikiANN-uz uses primarily Latin script: this is the most likely explanation for the 2% entity-string overlap reported in L2 (vs. $\sim 60\%$ for KZ/KY where script alignment is uniform Cyrillic). Since Uzbekistan officially uses Latin script and most contemporary formal text is Latin-based, our Uzbek perplexity and target-language NER results should not be generalized to Latin-script data without a separate evaluation. The script-mismatch nonetheless gives us an unintended benefit: Uzbek serves as a cleaner cross-lingual probe than KZ

or KY because memorization-driven inflation is negligible, so the strong transfer-induced UZ improvements (e.g., E1’s 0.578 F1) are difficult to attribute to data leakage.

N Functional Collapse: Raw Output Examples

To illustrate the nature of functional collapse at $r=64$, we present raw model outputs for two tasks. The model completely ignores instructions and exhibits degenerate repetition.

NER prompt: Extract named entities from the text. Format each as: ENTITY ||| TYPE

Input: Бишкекте Кыргыз Республикасынын Президентти Садыр Жапаров менен Казакстандын Президентти Касым-Жомарт Токаев жолукту.

$r=64$ output: Кыргыз Республикасынын Президентинин кызматына кандидат болгон. Кыргыз Республикасынын Президентинин кызматына кандидат болгон. Кыргыз Республикасынын Президентинин кызматына кандидат болгон... (repeats indefinitely)

Text continuation prompt: Кыргызстан --- это

$r=64$ output: не только, не только, не только, не только, не только... (single bigram repeated 25+ times until max tokens)

Both examples demonstrate that the model has lost instruction-following capability entirely. Rather than extracting entities or generating coherent text, it produces degenerate repetitions—a hallmark of functional collapse distinct from spectral collapse.

Experiment	PER F1	ORG F1	LOC F1
KZ Baseline 14.1M (on KY)	0.236	0.162	0.229
KZ Small-corpus 1.5M (on KY)	0.101	0.207	0.261
UZ Baseline (on KY)	0.095	0.150	0.209
KY Baseline	0.101	0.160	0.266
KY Overfit	0.224	0.255	0.119
KY $r=64$ (E5, lr 5×10^{-4})	0.000	0.000	0.000
KY $r=64$ (E5b, 5ep)	0.126	0.178	0.177
KY $r=64$ (E5c, 3ep)	0.107	0.139	0.114
KZ→KY (E6, related)	0.344	0.202	0.214
EN→KY (E6b, cross-family)	0.118	0.167	0.135

Table 14: Per-type NER F1 scores on Kyrgyz WikiANN. The KZ→KY transfer achieves the highest PER F1 (0.344), substantially outperforming the KY baseline (0.101). The $r=64$ configuration produces zero scores across all types due to 100% parse failures.

Experiment	KY	KZ	UZ
KZ Baseline (14.1M tok)	35.7%	38.4%	32.8%
KZ Small-corpus (1.5M tok)	37.3%	35.6%	32.8%
UZ Baseline	31.9%	33.1%	35.4%
KY Baseline	34.7%	30.2%	29.6%
KY Overfit	33.9%	27.8%	31.5%
KY $r=64$ (E5, lr 5×10^{-4})	23.2%	22.2%	22.9%
KY $r=64$ (E5b, 5ep)	33.2%	27.7%	26.6%
KY $r=64$ (E5c, 3ep)	38.5%	29.8%	31.9%
KZ→KY (E6, related)	33.7%	31.7%	30.8%
EN→KY (E6b, cross-family)	36.1%	29.1%	29.2%

Table 15: TUMLU accuracy across all experiments and languages. The KZ baseline achieves the highest accuracy on its target language (38.4%). The $r=64$ configuration drops to near-random levels ($\sim 23\%$ on 4-option multiple choice) across all languages.

O Robustness of the Pairwise-Cosine Result to Choice of “Final” Adapter

Table 16 reproduces the eleven principal pairwise comparisons from Table 3 using the literal last saved checkpoint of each run instead of the `final_adapter/` directory (which HF Trainer populates with the lowest-eval-loss checkpoint when `load_best_model_at_end=True`). All pairs move by at most ± 0.006 . The bimodal split between transfer-init pairs ($\cos \geq 0.34$) and direct-training pairs ($\cos < 0.07$) is preserved.

P Mechanistic Context

Three lines of prior work connect to our findings.

Loss-surface geometry and mode connectivity. Garipov et al. (2018) and Frankle et al. (2020) showed that independently trained networks often occupy distinct minima connected only by non-trivial paths, and that linear mode connectivity emerges only after a stability point in training. Our cross-seed cosines for direct LoRA fine-tuning

Pair	best-eval	lit. last	Δ
E3 cross-seed	0.041	0.043	+0.001
E5 cross-seed	0.022	0.022	+0.000
E5b cross-seed	0.030	0.030	+0.000
E5c cross-seed	0.032	0.033	+0.001
E8 cross-seed	0.014	0.016	+0.002
E1b cross-seed	0.028	0.032	+0.004
E6 cross-seed [†]	0.785	0.779	−0.006
E6b cross-seed	0.343	0.344	+0.001
E5 vs. E5c	0.010	0.010	+0.000
E5 vs. E5b	0.011	0.010	−0.001
E5b vs. E5c	0.053	0.049	−0.005

Table 16: Per-module mean cosine for eleven principal pairs, computed twice: against `final_adapter/` (best-eval, used in Table 3) and against the literal last saved checkpoint. The bimodal structure of Section 4.6 is robust to the choice. [†]Both E6 seeds warm-started from the same E1-s42 source; the independent-init version is 0.023 (Table 3).

(~ 0.02 – 0.04) are the LoRA-update analogue: two seeds of an identical configuration end up in essentially orthogonal regions of *BA* space, indicating that direct adapter optimisation does not enter a single shared basin. Transfer initialisation *from a shared source* (E6 with both seeds warm-started from E1-s42: $\cos = 0.785$, $294/294 > 0.5$) is the linearly-connected case in this analogy: starting from a non-trivial pinning point produces a single shared cone whose endpoint is reproducible. *Without* a shared source (independent E1-s42 and E1-s123: $\cos = 0.023$), the connected-cone property does not propagate – the source itself behaves like an independently-trained network.

Intrinsic dimensionality. Aghajanyan et al. (2021) showed that LLM fine-tuning operates in a very low intrinsic dimension—a small number of directions suffice to recover most of the task gain. A natural reading of our directional non-

canonicity is that the low intrinsic dimension is a property of the *loss landscape*, not of any specific axis: there are many low-dimensional subspaces that produce equivalent target-language fit, and direct LoRA selects among them stochastically through seed/initialisation. Which low-dim subspace is selected varies across seeds; the shape (singular-value distribution) is rank-biased and similar across runs, the spatial orientation is not.

LP-FT and feature distortion. Kumar et al. (2022) argued that full fine-tuning can distort pretrained features in ways that hurt OOD behaviour, and that linear-probe-then-fine-tune (LP-FT) mitigates this. Our cross-lingual forgetting under direct LoRA training is consistent with feature distortion: large $\|B\|_F$ growth on a Turkic adapter induces high cross-lingual PPL on other languages despite sub-threshold SE. Transfer-init functions as an LP-FT-like prior for the low-resource cross-lingual setting: a warm-start adapter has already paid most of the distortion cost on a related task, so the residual fine-tuning lives in a narrower cone and disturbs fewer pretrained directions. The graded E6 vs. E6b contrast (related-language vs. cross-family warm-start) shows the regularising effect scales with source-target similarity.

LoRA initialisation work. PiSSA (Meng et al., 2024) initialises LoRA factors from a truncated SVD of W_0 , biasing the adapter toward already-important pretrained directions; LoRA+ (Hayou et al., 2024) adjusts learning rates of A vs. B to address LoRA’s factorisation asymmetry. Both are direction-pinning interventions in our sense: they constrain where the optimisation can start (PiSSA) or how it can move (LoRA+). Our directional analysis suggests an empirical question for that line of work: *because PiSSA initialises both seeds from the same SVD-derived point* (the W_0 truncation is deterministic), it should sit in our shared-init regime; do the resulting cross-seed cosines reach E6’s shared-init level (~ 0.78), or only the partial regime of E6b (~ 0.34)? Independence-fair cross-seed protocols (different deterministic init schemes per seed) would be needed to test PiSSA against the independent-init regime (~ 0.02). The per-module pairwise-cosine protocol of Section 4.6 provides a direct test.

The empirical structural claim we add to this body of work is that within standard direct LoRA fine-tuning the directional non-canonicity is large enough to set the noise floor on any final-adapter-geometric diagnostic, and that the con-

nection between this noise floor and cross-lingual catastrophic-forgetting phenomenology is, to our knowledge, novel.

Q Additional Pairwise Cosines

Table 17 reports the remaining pairwise cosines we computed but moved out of the main-body Table 3 for space: cross-language $r=16$ comparisons, transfer-vs-direct, and other miscellaneous controls (including the BF16 quantisation contrast).

Pair	mean	med.	>0.5
<i>Cross-language $r=16$</i>			
E1 vs E3 (KZ/KY)	0.002	0.001	0/294
E1 vs E2 (KZ/UZ)	0.001	0.000	0/294
E3 vs E2 (KY/UZ)	0.002	0.001	0/294
<i>Transfer vs direct</i>			
E3 vs E6 (direct/KZ→KY)	0.009	0.005	0/294
E3 vs E6b (direct/EN→KY)	0.055	0.042	0/294
E6 vs E6b	0.007	0.005	0/294
<i>Other controls</i>			
E3 vs C3 (4-bit vs BF16)	0.066	0.053	0/294
E5 vs E3	0.005	0.003	0/294
E5 vs E1	0.001	0.001	0/294
E5 vs E8	0.000	0.000	0/294

Table 17: Additional pairwise cosines complementing Table 3. Cross-language pairs sit at ~ 0.001 – 0.002 (lower than same-language cross-seed direct training, 0.01 – 0.04). 4-bit vs. BF16 (E3 vs. C3) is 0.066 —an order of magnitude smaller than the transfer-init regime. Collapse vs. direct-baseline pairs (E5 vs. E1/E3/E8) all sit at 0.000 – 0.005 .

R Trajectory Cosines Across Checkpoints

PEFT saves intermediate checkpoints every ~ 1 epoch by default. For each seed-replicated configuration we compared two seeds at every shared checkpoint step. The cross-seed mean cosine is essentially flat across training (Table 18): direct-training pairs stay at 0.01 – 0.04 from the earliest available checkpoint through best-eval; transfer-init pairs under shared-source protocol stay at 0.34 – 0.78 across the same span (E6 and E6b second seeds both warm-started from the same E1-s42 / E8 sources). Seed-orthogonality, where it exists, is therefore established by step 1800 at the latest—earlier than we can measure—and the rest of training does not change the regime. The architecturally-fair independent-init test of Table 3 (E6-indep-s123 from E1-s123) breaks the high-cosine pattern of E6 ($0.785 \rightarrow 0.023$), confirming that the trajectory stability is conditional on a shared source rather than an intrinsic property of warm-start training.

Pair	step 1800	step 2800	best-eval
E3 cross-seed	0.034	—	0.041
E5 cross-seed	—	0.022	0.022
E5b cross-seed	0.030	0.030	0.030
E5c cross-seed	0.032	0.033	0.032
E6 cross-seed [†]	0.785	0.779	0.785
E6b cross-seed	0.343	0.344	0.343

Table 18: Cross-seed pairwise cosines across training checkpoints. Dashes indicate that no checkpoint was saved at that step for that run. Numbers move by at most ± 0.005 from earliest to latest; the bimodal regime is fixed by step 1800. [†]Both E6 seeds warm-started from the same E1-s42 source; the independent-init variant is 0.023 (Table 3).

Within-run rotation. For each run we additionally computed $\cos(\text{adapter at step } s, \text{adapter at best-eval})$ for every intermediate step. The cosines stay above 0.93 in all runs we measured: E5 step-2800 vs. step-4885 final= 0.938; E5c step-2800 vs. step-2931 final= 0.972; E6 step-2800 vs. step-2931 final= 0.993. Each adapter therefore moves through a tight cone in ΔW space, and the cross-seed orthogonality of the main result is not a function of when we sample the trajectory.

S Directional Geometry: Full Results

Cosine computation

Equation (1) is evaluated without forming the $\text{out} \times \text{in}$ product, via $\langle B_1 A_1, B_2 A_2 \rangle_F = \text{tr}(A_1^\top B_1^\top B_2 A_2)$, which needs only $r \times r$ intermediates.

α matters for direction more than rank does (C5 control). At identical $r/\text{lr}/\text{epochs}/\text{seed}/\text{data}$, $\cos(\text{E5c}, \text{C5}) = 0.072$, near the cross-seed band, but $\cos(\text{C5}, \text{E3}) = 0.155$ (2/294 modules above 0.5)—higher than any other direct-from-base pair. So $r=64$ at $\alpha=32$ (C5) sits directionally closer to the $r=16$ baseline (also $\alpha=32$) than to the $r=64$ E5c (which uses $\alpha=128$). Consistent with the functional finding (Section 4.1): α —via its role in effective per-step LR—is the variable that determines where the adapter lands.

Robustness. `final_adapter/` is the best-eval-loss checkpoint (`HF load_best_model_at_end=True`); re-running the eleven principal pairs against the literal last saved checkpoint changes each cosine by at most ± 0.006 (Section O). Seed-orthogonality is stable across training: cross-seed direct cosines stay in 0.02–0.04 from the earliest checkpoint through best-eval, and E6 stays at ~ 0.78

throughout (Section R).

Llama-3-8B replication. The L3 sweep (E3, E5c, E5 seed-replicated at $n=2$; plus the C5 absolute- α control and a fully-independent two-seed E6 KZ→KY transfer pipeline) reproduces all four paper-level patterns: $\max \text{SE} \leq 0.34$, L3-E5 NER F1= 0 with TypeAcc preserved at 0.65–0.80, the rank-only E5c effect and its cross-lingual cost reproduce, and the underdetermination inequality holds— $\cos(\text{L3-E5c-s42}, \text{L3-E5c-s123}) = 0.072$ (within-config) $> \cos(\text{L3-E5}, \text{L3-E5c}) = 0.029$ (between-config), mirroring Gemma’s $0.032 > 0.010$. Source-init pinning under transfer also replicates: $\cos(\text{L3-E6}, \text{L3-KZ-source}) = 0.804$, mirroring Gemma’s 0.797. The independent two-source transfer cross-seed on Llama-3 gives $\cos(\text{L3-E6-s42}, \text{L3-E6-s123}) = 0.060$ – the architecturally-fair counterpart of the new Gemma 0.023 test (Table 3, Section T).

T Llama-3-8B Replication

We replicate five of the critical Gemma-2-9B configurations on Llama-3-8B (Dubey et al., 2024) via the byte-identical NousResearch mirror of the gated Meta repository. Same data, same hyperparameters as the Gemma originals, same 4-bit NF4 + QLoRA setup, seed 42 (with E3, E5c additionally replicated at seed 123). Runs: **L3-E3** ($r=16$, $\alpha=32$, $\text{lr}=2 \times 10^{-4}$, 3 epochs); **L3-E5c** ($r=64$, $\alpha=128$, $\text{lr}=2 \times 10^{-4}$, 3 epochs); **L3-E5** ($r=64$, $\alpha=128$, $\text{lr}=5 \times 10^{-4}$, 5 epochs); **L3-C5** ($r=64$, absolute $\alpha=32$ – breaking the $\alpha=2r$ convention – $\text{lr}=2 \times 10^{-4}$, 3 epochs); and a two-stage transfer pipeline **L3-E6** (Kazakh pretrain at $r=16$, $\alpha=32$ on the 14.1M KZ corpus, then KZ→KY warm-start on the 4.4M KY corpus at the same hyperparameters).

All central claims of the paper replicate qualitatively (Table 19). (i) SE never reaches the 0.7 threshold (max SE: L3-E3 0.335, L3-E5c 0.137, L3-E5 0.138; cf. Gemma-2 0.322, 0.112, 0.058). (ii) L3-E5 exhibits the same functional-vs-spectral collapse dissociation: NER F1 (gen) = 0 on all three languages (parse failure rates 97/94/71% across KY/KZ/UZ), TUMLU drops to 24.7% (near 25% random), *but* log-likelihood NER TypeAcc is preserved or even higher than the baseline (0.649/0.741/0.796 vs. L3-E3 0.577/0.696/0.680). Entity knowledge survives; structured-output instruction-following collapses; SE does not see it. (iii) Cross-lingual forget-

ting in L3-E5 is severe (KZ PPL 145, UZ PPL 286 vs. L3-E3 30/54), in the same direction as Gemma though milder in absolute magnitude. (iv) The apparent rank-only target-language improvement of E5c over E3 reproduces on Llama-3 (3.63 vs. 4.32 KY PPL; 3.86 vs. 4.78 on Gemma) with the same cross-lingual cost (KZ 41 vs. 30 on Llama-3). (v) **The L3-C5 control closes the absolute- α story on Llama-3:** holding $\alpha=32$ at $r=64$ makes the adapter functionally indistinguishable from the $r=16$ baseline (L3-C5 KY PPL 4.21 vs. L3-E3 4.32; KZ PPL 29.9 vs. 30.4; UZ PPL 51.5 vs. 54.3), while L3-E5c’s apparent target-PPL gain (3.63) is bought at the cost of cross-lingual KZ PPL (41.1). The cost tracks absolute α , not rank—identical to the Gemma C5 result.

Directional underdetermination replicates, with quantitative scope. The critical pairwise cosines on Llama-3 (Table 20) preserve the qualitative regime structure of Gemma-2 but at slightly higher absolute values. Cross-configuration: $\cos(\text{L3-E5}, \text{L3-E5c}) = 0.029$, $\cos(\text{L3-E5}, \text{L3-E3}) = 0.015$, $\cos(\text{L3-E3}, \text{L3-E5c}) = 0.113$. Cross-seed: $\cos(\text{L3-E3-s42}, \text{L3-E3-s123}) = 0.103$, $\cos(\text{L3-E5c-s42}, \text{L3-E5c-s123}) = 0.072$. The Llama-3 cross-seed cosines are roughly $2\times$ higher than the Gemma counterparts (0.03–0.04), so direct LoRA training on Llama-3 is somewhat less seed-orthogonal than on Gemma. The underdetermination inequality of Section 5.2 is satisfied on this architecture too—the within-configuration cross-seed cosine (0.072 for E5c) is *larger* than the between-configuration cosine between collapse and healthy at the same rank (0.029 for E5 vs E5c), so no Lipschitz invariant can separate them, exactly as on Gemma. The C5 directional probe replicates: $\cos(\text{L3-C5}, \text{L3-E3}) = 0.343$ (same absolute α , different rank) vs. $\cos(\text{L3-E5c}, \text{L3-C5}) = 0.162$ (same rank, $4\times$ absolute α) – at matched absolute α the adapter occupies a near-shared direction regardless of rank, mirroring Gemma’s 0.155 vs. 0.072 contrast at $\sim 2\times$ scale. Seed-replicating C5 (new in this revision) brings Llama-3 to $n=2$ on all five configurations and exposes a boundary of the inequality worth stating plainly: $\cos(\text{C5-s42}, \text{C5-s123}) = 0.200$ is *below* the between-configuration $\cos(\text{C5}, \text{E3}) = 0.343$. For this pair the within-config spread does *not* exceed the between-config gap, so the inequality of Section 5.2 does not hold universally across configuration pairs—it holds for the collapse-vs-healthy comparisons the argument is about, on both architectures. The C5/E3 excep-

tion is itself consistent with the mechanism we report: two runs sharing an absolute α land in a common direction *across ranks* more reliably than two seeds of one configuration do, which is the strongest form of the “ α determines direction, rank does not” finding. Functionally C5 reproduces tightly across seeds (target KY PPL 4.21 vs. 4.19; TypeAcc 0.568 vs. 0.577), the same behaviour-vs-direction dissociation seen throughout.

Source-init pinning is architecture-independent; canonical convergence is not. The L3-E6 two-stage pipeline at seed 42 trains a Kazakh source-language baseline (L3-KZ-s42, KZ eval PPL 2.76) and then warm-starts on Kyrgyz under the same hyperparameters as L3-E3. Functionally, L3-E6 **retains Kazakh under Kyrgyz training:** KZ PPL drops from 30.38 (L3-E3 direct) to 19.04 (L3-E6 transfer), a 37% reduction (Gemma E6: $\sim 53\%$); target KY PPL is comparable (4.32 $\rightarrow$ 4.12); UZ degrades (54.3 $\rightarrow$ 81.6), consistent with the transfer originating from KZ rather than UZ. Directionally, L3-E6 is pinned to its source: $\cos(\text{L3-E6-s42}, \text{L3-KZ-s42-source}) = 0.804$ with 224/224 modules above 0.5, mirroring Gemma E6-s42 vs. E1-s42 source (0.797). **However, two transfers from independent sources do not share direction across seeds:** we ran the architecturally-fair test on both architectures – L3-E6-s42-from-L3-KZ-s42 vs. L3-E6-s123-from-L3-KZ-s123 gives $\cos = 0.060$ (1/224 above 0.5); the Gemma counterpart (E6-s42-from-E1-s42 vs. E6-indep-s123-from-E1-s123, run in this revision) gives $\cos = 0.023$ (0/294). Both numbers sit inside the direct-training cross-seed band of Section 4.6. The previously-reported Gemma cross-seed cosine 0.785 reflects two stage-2 warm-starts from a *shared* E1-s42 source (the only KZ baseline available at submission time); it is essentially a measure of source-init pinning, not cross-seed canonicity. *Functionally*, transfer retention reproduces across architectures and across independent reseeding (L3-E6-s123 from L3-KZ-s123: KY 4.17, KZ 18.5; Gemma E6-indep-s123 from E1-s123: KY 4.58, **KZ 26.79 vs direct training 50.0 ± 2.4 , a -47% retention effect that matches the original shared-init E6 within seed spread**). *Directionally*, transfer pins to its source but does not produce a canonical direction – the underdetermination argument of Section 5.2 extends to transfer once the source is reseeded.

	E3		E5c		E5		C5	
	L3	G2	L3	G2	L3	G2	L3	G2
KY PPL	4.32	4.78	3.63	3.86	3.43	5.90	4.21	4.46
KZ PPL (x-ling)	30.4	47.6	41.1	93.5	145.1	659.3	29.9	48.9
UZ PPL (x-ling)	54.3	56.9	80.1	111.3	285.6	742.7	51.5	60.0
NER F1 KY (gen)	0.143	0.157	0.165	0.115	0.000	0.000	0.144	—
NER TypeAcc KY	0.577	0.504	0.531	0.549	0.649	0.595	0.568	—
TUMLU KY	32.3%	34.7%	31.9%	38.5%	24.7%	23.2%	31.1%	—
max SE	0.335	0.322	0.137	0.112	0.138	0.058	0.205	—

Table 19: Llama-3-8B (L3) vs Gemma-2-9B (G2) on the four replicated configurations. All paper-level patterns reproduce: E3 healthy baseline, E5c better target with cross-lingual cost, E5 catastrophic functional collapse at sub-threshold SE with NER F1=0 on all three languages, entity knowledge preserved under log-likelihood TypeAcc, near-random TUMLU. **The L3-C5 column closes the absolute- α story on Llama-3:** at $r=64$ with $\alpha=32$ (breaking the $\alpha=2r$ convention), the adapter is functionally indistinguishable from the $r=16$ baseline across all metrics; the apparent E5c r -only benefit was an artefact of the convention amplifying absolute α at higher rank. Magnitudes of cross-lingual forgetting are milder on Llama-3 than on Gemma, but direction is identical.

Functional consistency across L3 seeds. Table 21 shows that the directional spread across seeds is not matched by a similar spread in downstream functional metrics: target PPL stays within ± 0.05 across seeds for both E3 and E5c, TUMLU within ± 1.5 percentage points, and TypeAcc within ± 0.03 . This is the same behaviour-vs-direction dissociation we observe on Gemma: two seeds of the same configuration produce essentially identical model behaviour through essentially orthogonal weight-space directions.

Pair	L3	G2
<i>Cross-config, same seed</i>		
cos(E5, E5c) – collapse vs. healthy ($r=64$)	0.029	0.010
cos(E5, E3) – collapse vs. $r=16$ baseline	0.015	0.005
cos(E3, E5c) – $r=16$ vs. $r=64$	0.113	—
<i>Cross-seed, same config (direct training)</i>		
cos(E3-s42, E3-s123) – $r=16$	0.103	0.041
cos(E5c-s42, E5c-s123) – $r=64$ healthy	0.072	0.032
cos(E5-s42, E5-s123) – $r=64$ collapse	0.037	0.022
<i>Absolute-α control (C5)</i>		
cos(C5, E3) – same abs α , diff. rank	0.343	0.155
cos(E5c, C5) – same rank, $4\times$ abs α	0.162	0.072
cos(C5-s42, C5-s123) – <i>cross-seed</i>	0.200	—
<i>Transfer (E6): source-pinning vs. canonicity</i>		
cos(E6, its KZ source)	0.804	0.797
cos(E6, E3) – transfer vs. direct KY	0.020	0.009
cos(E6-s42, E6-s123, <i>shared</i> init)	—	0.785
cos(E6-s42, E6-s123, <i>independent</i> inits)	0.060	0.023

Table 20: Per-module pairwise cosine on Llama-3-8B (L3) vs. Gemma-2-9B (G2). (i) The collapse-vs-healthy comparison cos(E5, E5c) sits in the same regime on both architectures (≤ 0.03 , 0 modules above 0.5). (ii) Cross-seed direct-training cosines on L3 (0.07–0.10) are $\sim 2\times$ G2’s (0.03–0.04); same scaling for the C5 probe (0.343 vs. 0.155). (iii) **Transfer pins to source, not to a canonical direction.** The adapter is pinned to its own KZ source on both architectures (cos ≈ 0.80). The headline Gemma cross-seed 0.785 was measured with two stage-2 warm-starts from a *shared* E1-s42 source (E1-s123 did not exist at submission); the architecturally-fair test – two transfers from *independent* KZ baselines – gives 0.060 on L3 and 0.023 on Gemma, inside the direct-training band. Functional retention reproduces under independent reseeding on both architectures; directional convergence does not. The underdetermination inequality of Section 5.2 thus extends to transfer once the source is reseeded.

Metric	L3-E3 (s42 / s123)	L3-E5c (s42 / s123)
KY PPL	4.32 / 4.36	3.63 / 3.68
KZ PPL (cross-ling.)	30.4 / 32.1	41.1 / 45.7
UZ PPL (cross-ling.)	54.3 / 55.1	80.1 / 75.9
NER F1 KY (gen)	0.143 / 0.200	0.165 / 0.151
NER TypeAcc KY	0.577 / 0.549	0.531 / 0.531
TUMLU KY	32.3% / 30.8%	31.9% / 31.5%

Table 21: Llama-3-8B cross-seed functional metrics for the two seed-replicated direct-training configurations. Functional metrics are tight across seeds (KY PPL within ± 0.05 , TUMLU within ± 1.5 pp), even though the per-module direction differs substantially (cos E3 = 0.103, E5c = 0.072; Table 20). Same behaviour-vs-direction dissociation as Gemma-2: similar functional outcomes through essentially orthogonal weight-space directions.